
PINNfluence: Interpreting PINNs through Influence Functions

Aleksander Krasowski^{*1} Jonas Naujoks^{*1} Moritz Weckbecker¹ Galip Ü. Yolcu¹ Thomas Wiegand¹
Sebastian Lapuschkin¹ Wojciech Samek¹ René P. Klausen¹

Abstract

Physics-informed neural networks (PINNs) have emerged as a powerful deep learning approach for solving partial differential equations (PDEs) in the physical sciences, yet their behavior remains largely opaque and is typically understood through failure mode analyses rather than explicit interpretability. To address this issue, we introduce PINNfluence, a training data attribution framework for interpreting PINNs based on influence functions. By extending influence functions to composite physics-informed training objectives, we enable fine-grained attribution between predictions, loss components, and training data points. Through benchmark experiments across various PDEs, we demonstrate that influence patterns provide granular diagnostics that distinguish different PINN failure modes. PINNfluence thus opens a new avenue for understanding and improving the reliability of PINNs through the lens of their data.

1. Introduction

Partial differential equations (PDEs) are the central modeling language in the sciences, describing how quantities evolve across space and time under governing laws. Solving these equations accurately and efficiently is essential for scientific prediction and control, yet often computationally demanding. In recent years, machine learning (ML) approaches have proven themselves exceptionally adept at approximating solutions to such problems, bundling fast inference and strong empirical performance once trained. Physics-informed neural networks (PINNs) (Raissi et al., 2019) integrate physical knowledge by embedding PDEs into neural network training objectives (Karniadakis et al.,

2021; Kim et al., 2021; Cuomo et al., 2022), enabling applications across fluid mechanics, electromagnetics, disease modeling, and optics, inter alia (Cai et al., 2021; Beltrán-Pulido et al., 2022; Berkahn & Ehrhardt, 2022; Medvedev et al., 2023). Despite their flexibility, understanding and improving PINN behavior when training fails remains challenging (Wang et al., 2021; Krishnapriyan et al., 2021; Wang et al., 2022; Rathore et al., 2024; Liu et al., 2025).

The existing understanding of PINNs is largely based on analyses of training dynamics, failure modes, and ill-conditioned loss landscapes. Prior work attempts to alleviate these issues through improvements in optimization, architecture, objective function design, and training data selection (Jagtap et al., 2020; McClenny & Braga-Neto, 2023; Wang et al., 2023; Wu et al., 2025; Jeong et al., 2025). Further, rigorous error guarantees and uncertainty quantification for PINNs are active fields of research and often difficult to obtain in practice (Eiras et al., 2024; Hillebrecht & Unger, 2025). Both of these strands of research aim to make PINNs reliable and performant, qualities that are crucial for wide-scale adoption. In parallel, a substantial body of work on explainable artificial intelligence (XAI) has developed tools to interpret and diagnose complex machine learning models, improving trust, scientific insights, and model validation (Ross et al., 2017; Rieger et al., 2020) in varied domains such as medical imaging (Hense et al., 2024), computer vision (Bach et al., 2015; Bender et al., 2023) and natural language processing (Bricken et al., 2023).

While interpretability¹ has become a central theme in deep learning (Samek et al., 2021; Rudin et al., 2021) and scientific machine learning in particular (Wetzel et al., 2025), PINNs still lack dedicated explainability methods that support transparency, systematic diagnosis and validation – a precondition for their widespread application. In particular, although existing analyses of PINN failure modes provide valuable insights at the level of training dynamics and model classes, they fall short of providing diagnostic or attribution-based explanations that reveal how individual training points and physics constraints shape model predictions. This leaves a gap for training-data-centric interpretability methods for PINNs, which we address in this work. Such methods have

¹Department of Artificial Intelligence, Fraunhofer HHI, Berlin, Germany. Correspondence to: Aleksander Krasowski <aleksander.krasowski@hhi.fraunhofer.de>, Jonas Naujoks <jonas.naujoks@hhi.fraunhofer.de>, René P. Klausen <rene.pascal.klausen@hhi.fraunhofer.de>.

¹We use this term interchangeably with explainability.

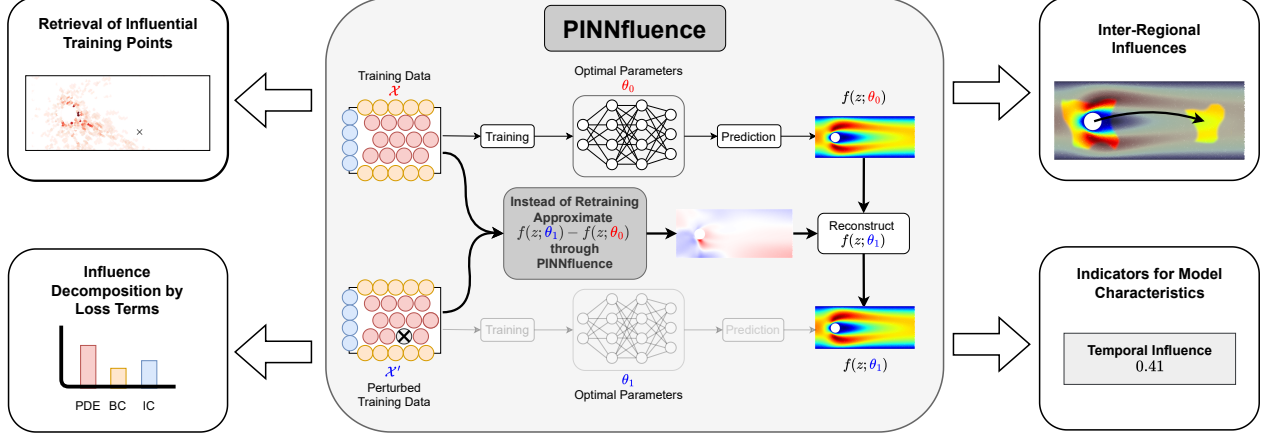


Figure 1. PINNfluence approximates the effect of perturbing training data by estimating the difference $f(z, \theta_1) - f(z, \theta_0)$ without retraining. Here, θ_0 denotes the parameters obtained from the original training set $\mathcal{X}$, while θ_1 corresponds to a perturbed training set $\mathcal{X}'$ that differs by a single training point. This enables attribution of model predictions at test points z to individual training samples and loss components, supporting (i) retrieval of influential training points for the given prediction, (ii) inspection of how one region influences another, (iii) decomposition by individual loss terms, and (iv) construction of diagnostic indicators that characterize model behavior.

the potential to improve transparency and robustness by enabling fine-grained diagnosis and control of PINN behavior, thereby complementing existing approaches.

Specifically, we adapt influence functions, a training data attribution (TDA) method, to PINNs. TDA quantifies the influence of individual training points on model behavior and has proven effective for debugging models, identifying memorization and harmful data, and understanding model behavior through the lens of the training set in domains such as computer vision and large language models (Koh & Liang, 2017; Pruthi et al., 2020; Braun et al., 2022; Hamoudeh & Lowd, 2024). An overview of our proposed PINNfluence framework and the types of influence-based analyses it provides is shown in Figure 1.

Our contributions can be summarized as follows:

- We generalize influence functions to PINNs, resulting in PINNfluence, an interpretability framework that attributes predictions to individual training points and decomposes their influence by loss components.
- We provide a toolbox of diagnostic indicators, including temporal, directional, and region-to-region measures, that enable systematic analysis of trained PINNs.
- Across five time-dependent PDEs, we identify consistent structural differences between well-trained and poorly-trained models. Well-trained models show diminishing influence of the initial condition (IC), while poorly-trained models exhibit persistent IC dependence over time.

2. Theoretical Background

2.1. Physics-Informed Neural Networks (PINNs)

PDEs are ubiquitous modeling tools in the computational sciences: For example, they describe the motion of galaxies or the spread of infectious diseases. A PDE is typically posed together with initial and boundary conditions, such as the starting position of galaxies or the initial amount of infected specimens. A PDE with these constraints is called an initial-boundary value problem (IBVP). PINNs (Raissi et al., 2019) provide a widely-used ML-based framework for approximating solutions of IBVPs. Let $\Omega \subset \mathbb{R}^n$ be a connected, bounded, open set with piecewise smooth boundary $\partial\Omega$, and define its closure by $\bar{\Omega} := \Omega \cup \partial\Omega$. We study PDE constraints in the domain Ω together with K boundary and/or initial constraints posed on subsets $\Gamma_k \subseteq \partial\Omega$:

$$\mathcal{N}[u](x) = 0, \quad x \in \Omega \quad (1)$$

$$\mathcal{B}_k[u](x) = 0, \quad x \in \Gamma_k \text{ for } k = 1, \dots, K, \quad (2)$$

where $\mathcal{N}$ and $\mathcal{B}_k$ are differential operators acting on the solution $u : \bar{\Omega} \rightarrow \mathbb{R}^d$ of (1) and (2) with K initial and/or boundary conditions. Here, x denotes the spatio-temporal coordinates, which may be purely spatial, purely temporal, both, or more generally any physical quantity on which the PDE depends. The above formulation (2) encompasses both Dirichlet and Neumann boundary conditions, among others. The aim of the PINN approach is to approximate a solution $u(x)$ of (1) and (2) with a neural network $\phi(x; \theta)$ by optimizing its trainable parameters $\theta \in \Theta \subseteq \mathbb{R}^p$. Training proceeds by minimizing an empirical risk defined through

the composite loss function

$$\mathcal{L}(\theta) := \frac{\lambda_{\text{pde}}}{|\mathcal{X}_{\text{pde}}|} \sum_{x \in \mathcal{X}_{\text{pde}}} L_{\text{pde}}(x; \theta) + \sum_{k=1}^K \frac{\lambda_{\text{bc},k}}{|\mathcal{X}_{\text{bc},k}|} \sum_{x \in \mathcal{X}_{\text{bc},k}} L_{\text{bc},k}(x; \theta), \quad (3)$$

with per-sample losses $L_{\text{pde}}(x; \theta) := \|\mathcal{N}[\phi(x; \theta)]\|_2^2$, $L_{\text{bc},k}(x; \theta) := \|\mathcal{B}_k[\phi(x; \theta)]\|_2^2$ and weights $\lambda_{\text{pde}}, \lambda_{\text{bc},k} > 0$. We consider training points $\mathcal{X}_{\text{pde}} \subset \Omega$ and $\mathcal{X}_{\text{bc},k} \subset \Gamma_k$, which construct $\mathcal{X}_{\text{train}} = \mathcal{X}_{\text{pde}} \cup \bigcup_k \mathcal{X}_{\text{bc},k}$. Data-driven regression terms for incorporating measurements can be included but are omitted here for the sake of clarity.

2.2. Influence Functions (IFs)

Influence functions (IFs) (Hampel, 1974; Cook & Weisberg, 1982; Koh & Liang, 2017) address the question of how a model’s behavior would change if specific training points were removed from or added to the training set. Through a linear approximation of leave-one-out retraining, IFs enable the analysis of a model in terms of the contributions of individual training points. To adapt this framework to PINNs, we generalize the approach of (Koh & Liang, 2017; Koh et al., 2019) to composite loss functions arising from physics-based constraints.

Definition 2.1 (Influence Function). Let $L : \bar{\Omega} \times \Theta \rightarrow \mathbb{R}_{\geq 0}$ be a twice differentiable, single-sample loss function, that induces an empirical risk $\mathcal{L}(\theta) = \frac{1}{N} \sum_{x \in \mathcal{X}} L(x; \theta)$ over a training set $\mathcal{X} \subset \bar{\Omega}$ of size N , and $f : A \times \Theta \rightarrow \mathbb{R}^m$ be an arbitrary differentiable function. For two points $x \in \bar{\Omega}$, $z \in A$, we call

$$\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) := -\nabla_{\theta} f(z; \theta_0)^{\top} \mathcal{H}_{\theta_0}^{-1} \nabla_{\theta} L(x; \theta_0) \quad (4)$$

the *influence function* from L at the point x towards f at the point z with respect to θ_0 , where $\mathcal{H}_{\theta_0} = \nabla_{\theta}^2 \mathcal{L}(\theta_0)$ denotes the Hessian at θ_0 , which is assumed to be invertible.

Theorem 2.2 (Influence Functions Approximate Retraining). Assume the setting of Definition 2.1 with L and f being analytic. Further, assume that θ_0 is a non-degenerate stationary point of $\mathcal{L}$ with respect to a training set $\mathcal{X}$ (i.e., $\nabla_{\theta} \mathcal{L}(\theta_0) = 0$, $\det \mathcal{H}_{\theta_0} \neq 0$). Let θ_1 denote the stationary point of $\mathcal{L}(\theta)$ with respect to $\mathcal{X}'$, where we added or removed a training point $x^{\pm}$. For sufficiently large N , the resulting effect on the function f is given by

$$f(z; \theta_1) - f(z; \theta_0) = \pm \text{Inf}_{\theta_0}^{L \rightarrow f}(x^{\pm}, z) \frac{1}{N} + \mathcal{O}(1/N^2) \quad .$$

We provide the proof in Appendix A.1. Thus, the IF computes the effect of infinitesimally upweighting the contribution of a data point in the training objective. Downweighting a training point yields a sign-flipped influence expression.

In practice, IFs are most often evaluated for the influence on the test loss. This corresponds to the special case $f = L$, which reads $\text{Inf}_{\theta_0}^{L \rightarrow L}(x, z)$. It is important to note that IFs constitute only a first-order approximation. In practical application, discrepancies may arise when the number of training points N is not sufficiently large or when θ_0 is only close to being a stationary point. Further, note that the accuracy of the approximation depends on the local geometry of the loss landscape.

Compared to the original formulation of influence functions (Koh & Liang, 2017), we generalize the framework in several important ways. First, we extend the analysis from strict minima to non-degenerate stationary points, which is critical in neural network training, where optimization typically converges to saddle points rather than global optima (Dauphin et al., 2014). This relaxation is especially relevant in PINNs, whose optimization landscapes are often ill-conditioned (Rathore et al., 2024; Ryck et al., 2024). Second, we refine the required assumptions on the loss landscape, noting that strong convexity, while necessary, is not sufficient. Finally, by allowing f to be an arbitrary differentiable function, we generalize IFs from loss-to-loss attributions to influence from any quantity to any component of a composite objective, which is crucial for analyzing PINN objectives.

Corollary 2.3 (Properties of Influence Functions). Assume the setting of Definition 2.1 and Theorem 2.2.

(Linearity) The influence function is bilinear in the loss L and the quantity of interest f . For scalars $\alpha, \beta \in \mathbb{R}$, and suppressing the dependence on (x, z) for brevity, we have

$$\begin{aligned} \text{Inf}_{\theta_0}^{\alpha L_1 + \beta L_2 \rightarrow f} &= \alpha \text{Inf}_{\theta_0}^{L_1 \rightarrow f} + \beta \text{Inf}_{\theta_0}^{L_2 \rightarrow f}, \\ \text{Inf}_{\theta_0}^{L \rightarrow \alpha f_1 + \beta f_2} &= \alpha \text{Inf}_{\theta_0}^{L \rightarrow f_1} + \beta \text{Inf}_{\theta_0}^{L \rightarrow f_2} \quad . \end{aligned}$$

(Additivity) In addition, we can remove or add several points. Let $\mathcal{X}^+$ be a set of points added to the training set and $\mathcal{X}^- \subset \mathcal{X}$ a set of points removed. Then, for sufficiently large N , the resulting change in f satisfies

$$N(f(z; \theta_1) - f(z; \theta_0)) \approx \sum_{x \in \mathcal{X}^+} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) - \sum_{x \in \mathcal{X}^-} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) \quad .$$

This corollary enables the analysis of influence between arbitrary regions of the domain and individual components of the PINN objective, forming the basis of the PINN-FLUENCE framework developed in Section 4.

Beyond point-wise effects, it is often practical to consider aggregated measures such as the total influence $\sum_{z \in R} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)$ or its absolute variant $\sum_{z \in R} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|$ over regions $R \subseteq \bar{\Omega}$. Together with

Corollary 2.3, this permits the study of influences from points to regions, regions to points, and region-to-region interactions, while simultaneously allowing for the analysis of individual loss components. For a discussion on the practical considerations of the computation of influences, such as circumventing the costly explicit construction of the Hessian, we refer to Appendix A.2.

3. Interpretability Approaches for PINNs and Related Work

Interpretability and explainability have emerged as central themes in the machine learning community, driven by the increasing deployment of complex models in high-stakes applications. Naturally, this trend has also extended to scientific machine learning (Schnake et al., 2022; Kitouni et al., 2024; Wetzel et al., 2025), where model transparency and verification are closely tied to scientific validity. Still, dedicated interpretability methods for PINNs remain scarce.

Failure Modes of PINNs In the absence of dedicated interpretability frameworks, much of the existing understanding of PINNs has been developed through the systematic analysis of their failure modes. There is a significant line of work investigating the different mechanisms underlying these behaviors. This includes gradient flow pathologies and ill-conditioned loss landscapes (Wang et al., 2022; Rathore et al., 2024; Ryck et al., 2024), information propagation failure (Krishnapriyan et al., 2021; Daw et al., 2023), failure of initial condition learning (Steger et al., 2022), and different phases of learning in PINNs (Anagnostopoulos et al., 2024). Substantial efforts have focused on mitigating these failures, including loss reweighting (Wang et al., 2023; 2024), adaptive sampling (Lu et al., 2021; Liu et al., 2022; Wu et al., 2023), and refined network architectures (Jagtap et al., 2020; McClenny & Braga-Neto, 2023). Although these studies provide valuable diagnostic insights into training dynamics and optimization pathologies, they do not offer explicit post-hoc interpretability or attribution mechanisms.

Interpretability in PINNs and Learning of the Solutions of PDEs Another line of research aims to increase interpretability in scientific machine learning for PDEs more explicitly. Here, some approaches include constraining the architecture to increase interpretability (Ramabathiran & Ramachandran, 2021) or developing novel architectures to be interpretable by design, such as Kolmogorov-Arnold Networks (Liu et al., 2024) and GINN-KANs (Ranasinghe et al., 2024), but they do not address post-hoc analysis of already trained PINNs.

Although not aimed at PINNs, another emerging area of interpretability research is concerned with the post-hoc analysis of the learned representation of data-driven PDE

Table 1. Overview of PINNfluence-based diagnostics used in this work and the insights they provide.

Diagnostic Question	PINNfluence tool	Insight
Which training points shape a given prediction?	Point-wise influence maps	Localizes sensitivity to individual collocation points.
How do individual physics constraints shape model behavior?	Loss-component influence decomposition (6)	Quantifies the relative contribution of PDE, boundary, and IC. Exposes IC-dominated failure mode.
Do learned solutions reflect temporally oriented influence?	Temporal influence indicator (9)	Shows that well-trained PINNs exhibit near-symmetric temporal influence, while asymmetry signals failure.
How does influence distribute across domain subsets?	Regional influence indicators	Probes problem-specific structure: downstream propagation, spatial symmetry, shock locality, see Appendix C.

solvers. Recently, methods from the field of mechanistic interpretability (Bricken et al., 2023) such as sparse autoencoders (Cunningham et al., 2023) and linear probing (Alain & Bengio, 2018) have been used to investigate neural operators trained to solve PDEs and predict weather dynamics (MacMillan & Ouellette, 2025; Fear et al., 2025), discovering concept activation vectors and latent features that correspond to weather and flow phenomena such as vorticity and diffusivity. Although promising, these methods do not extend to PINNs.

Despite these advances, explicit attribution and diagnostic tools for PINNs remain largely unexplored. In particular, to the best of our knowledge, no training-data-centric framework for PINNs exists that enables attributing model behavior to individual collocation points or physics constraints, which motivates the approach proposed in this work.

4. Methodology of PINNfluence

In this section, we introduce our toolbox of PINNfluence-based analysis primitives and indicators for diagnosing PINN behavior. Conceptually, PINNfluence decomposes influence along two axes: *what* contributes and *where* contributions arise. Building on the theoretical results of Section 2, we show how PINNfluence can be utilized at different levels of granularity. Starting from point-wise influences, we construct decompositions of loss components and regions, and define normalized indicators that summarize influence effects for systematic comparison across models.

These tools are designed to test concrete diagnostic hypotheses about the behavior of an optimized model, such as “Which constraints dominate a prediction?” and “How does influence align with temporal or directional structure of the underlying PDE?” In Table 1, we give a summary of the concrete questions together with the insights they yield.

Point-wise Influences As fundamental building blocks of our analysis, point-wise influences (4) quantify how an individual training point affects a function f , such as the prediction, the loss or a physical observable, evaluated at a specific test location. Influence values can be positive and negative, indicating whether a training point increases or decreases f at the test location. Empirically, we observe that the signs of influence effects can be noisy. Since we are primarily interested in the strength of the effect, we mainly consider absolute influence values $|\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|$, which provide a more robust estimate of sensitivity. Further, points with large absolute influences can be interpreted as regions of high complexity for the model (Zhang & Zhang, 2022). Thus, point-wise influences are particularly suited for attributing model behavior at individual points in the domain Ω to specific training points. This enables diagnosis beyond localization: one can identify not only where a model fails, but also which training points dominate predictions across different regions, yielding a structural characterization invisible to residual analysis.

Point-to-Region Influences Building on the linearity and additivity properties of influence functions (see Corollary 2.3), we aggregate point-wise influences to study region-level effects. Throughout the section, we denote $R_{\text{tr}}, R_{\text{te}} \subset \Omega$ as subsets of training and test samples, respectively. Fixing a training point $x \in \mathcal{X}$ and summing its influence over test locations yields point-to-region influence maps, revealing where and how strongly this datum affects a quantity of interest across the domain.

$$\text{Inf}_{\theta_0}^{L \rightarrow f}(x, R_{\text{te}}) = \sum_{z \in R_{\text{te}}} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) \quad . \quad (5)$$

Notably, this provides insights into how the quantity of interest f would change at a macroscopic scale if the training point were perturbed or removed. Analogously, one can fix a test point and aggregate the estimated effects of a set of training points R_{tr} , which reveals how each training point contributes to model behavior at that location.

Decomposition by Individual Loss Components The linearity of the influence in the loss argument (Corollary 2.3) allows us to decompose it with respect to every individual loss term. This allows for fine-grained analysis of the effect of the PDE residual, or any of the boundary and initial conditions on a given test point’s prediction. Notably, residual inspection alone does not offer the same level of granularity, as it does not permit the attribution of predictions or quantities of interest to specific loss components or constraints. We obtain the *fractional contribution* of each loss term by taking its ratio with respect to the sum of all

absolute contributions, which yields

$$r_{L_i}(x, z) = \frac{|\text{Inf}_{\theta_0}^{L_i \rightarrow f}(x, z)|}{\sum_j |\text{Inf}_{\theta_0}^{L_j \rightarrow f}(x, z)|} \in [0, 1] \quad , \quad (6)$$

satisfying $\sum_i r_{L_i}(x, z) = 1$. Since individual influences may have opposing signs, we additionally report a cancellation score

$$\begin{aligned} \kappa(x, z) &= 1 - \frac{\left| \sum_j \text{Inf}_{\theta_0}^{L_j \rightarrow f}(x, z) \right|}{\sum_j |\text{Inf}_{\theta_0}^{L_j \rightarrow f}(x, z)|} \\ &= 1 - \frac{|\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}{\sum_j |\text{Inf}_{\theta_0}^{L_j \rightarrow f}(x, z)|} \in [0, 1] \end{aligned} \quad (7)$$

that quantifies the degree to which loss terms reinforce ($\kappa \approx 0$) or counteract ($\kappa \approx 1$) each other. When cancellation is low, the fractions reliably indicate each term’s contribution to the total influence. Conversely, high cancellation corresponds to competing constraints whose individual effects largely negate each other at the prediction level.

Region-based Indicators In analogous fashion to the loss components decomposition, we can decompose influence with respect to training and test regions. We first define relative regional influences, which capture the estimated effect on model behavior of certain regions relative to the whole domain. The *relative absolute regional influence* of R_{tr} on R_{te} is given by

$$\rho_{\theta_0}^{L \rightarrow f}(R_{\text{tr}}, R_{\text{te}}) = \frac{1}{|R_{\text{te}}|} \sum_{z \in R_{\text{te}}} \frac{\sum_{x \in R_{\text{tr}}} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}{\sum_{x \in \mathcal{X}_{\text{train}}} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|} \quad , \quad (8)$$

with $\rho_{\theta_0}^{L \rightarrow f} \in [0, 1]$. It allows quantifying how strongly a given training region contributes to model behavior over a specific test region. For region-level analysis, point-wise loss decompositions (6) can also be aggregated over subsets of training and test points $\bar{r}_{L_i}(R_{\text{tr}}, R_{\text{te}}) = \frac{1}{|R_{\text{te}}|} \sum_{z \in R_{\text{te}}} \sum_{x \in R_{\text{tr}}} r_{L_i}(x, z)$. An important variation of the aforementioned regional indicator arises in time-dependent problems. For such settings, we define a *temporal influence indicator* that quantifies how strongly the model behavior at test points is influenced by the training points in the past

$$\begin{aligned} \eta_{\theta_0}^{L \rightarrow f}(R_{\text{tr}}, R_{\text{te}}) &= \\ 1 - \frac{1}{|R_{\text{te}}|} \sum_{z \in R_{\text{te}}} \frac{\sum_{x \in R_{\text{tr}}: x_t \leq z_t} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}{\sum_{x \in \mathcal{X}_{\text{train}}} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|} \end{aligned} \quad (9)$$

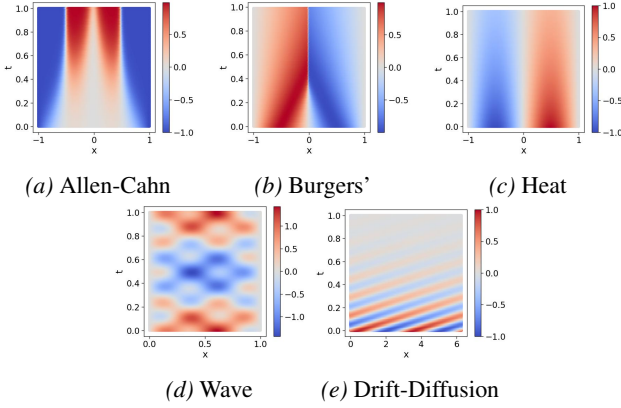


Figure 2. Investigated problems.

Again, we have $\eta_{\theta_0}^{L \rightarrow f} \in [0, 1]$. The subtraction from 1 ensures that smaller values of η correspond to stronger past-directed influence; larger values imply the opposite. Assuming that the distribution of training points is uniform, values of ≈ 0.5 are expected if influence is temporally symmetric. Notably, this notion cannot be probed through residual analysis alone. These indicators can be specialized to various other diagnostic hypotheses, e.g., boundary dominance, spatial locality, or symmetry, through appropriate choices of the loss component L_i , the function f , and the sets of training and test regions. In analogy to (9), directional indicators can be constructed for time-independent problems by defining ordering relations over spatial directions. For this and further instances, we refer to [Appendix C](#).

Indicators summarize influence pattern analysis and provide first-order, Hessian-based sensitivity diagnostics under the trained parameters of a PINN. Notably, PINNfluence does not provide formal error guarantees or interventional causal claims. It characterizes the converged model’s sensitivity to how predictions would respond to perturbations in the training data, rather than the training dynamics that produced the structure. In this sense, PINNfluence supports the comparative and hypothesis-driven diagnosis of trained PINNs, as demonstrated in the following section.

5. Experiments

We demonstrate how PINNfluence helps to understand the behavior of PINNs across five time-dependent PDEs spanning diverse physics: the heat, Allen-Cahn, Burgers’, wave, and drift-diffusion equations, shown in [Figure 2](#). Additionally, we provide an analysis of two steady-state problems, namely the Poisson and Navier-Stokes equations in [Appendix C.1](#). For every problem, we include two different settings: a *well-trained* model that accurately solves the IBVP, and a *poorly-trained* model that fails to accurately solve the IBVP due to insufficient data or suboptimal optimization. Comparing influence patterns between these

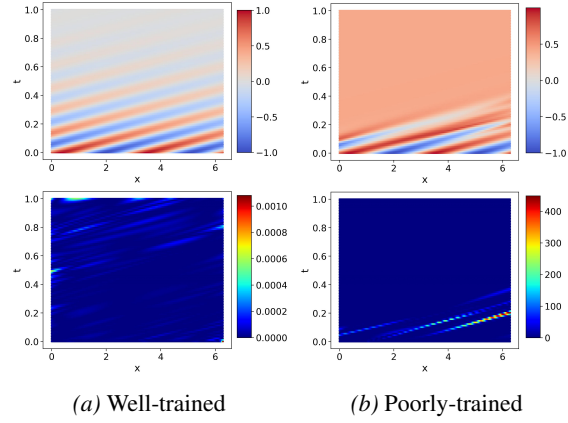


Figure 3. Prediction and losses for the drift-diffusion problem for a representative seed.

two configurations allows us to identify differences in how the two classes of models depend on their training data at convergence. This enables us to provide diagnostic indicators that distinguish well-performing from pathological model states in a controlled way. To assess consistency, we include 10 random seeds for each configuration. The complete problem specifications and training setup are provided in [Appendix B](#).

Across experiments, we focus on three recurring research questions that exemplify how PINNfluence yields insights into PINN behavior:

- (i) Can PINNfluence give insights into differences between well-trained and poorly-trained models, beyond localizing where errors occur?
- (ii) Can PINNfluence attribute model sensitivity to specific loss components L_i ?
- (iii) Can we reveal structural properties of trained PINNs using PINNfluence, such as temporal influence patterns that are not evident from residuals alone?

Point-wise Attribution To demonstrate the capabilities of PINNfluence, we focus our analysis on the drift-diffusion equation, a convection problem where PINNs are known to suffer from *propagation failure*, i.e., collapsing to near-constant solutions forward in time ([Krishnapriyan et al., 2021](#)). This exact failure mode is present in our poorly-trained configuration and is straightforward to localize. Predictions show the collapse to a constant output emerging at around $t = 0.2$ and the losses indicate a large PDE violation at the transition region, as shown in [Figure 3](#).

PINNfluence aids us in understanding how the prediction was shaped, through the lens of the PINN’s training data. [Figure 4](#) shows point-to-region influence attribution maps for model predictions at times $t > 0.5$, identifying training

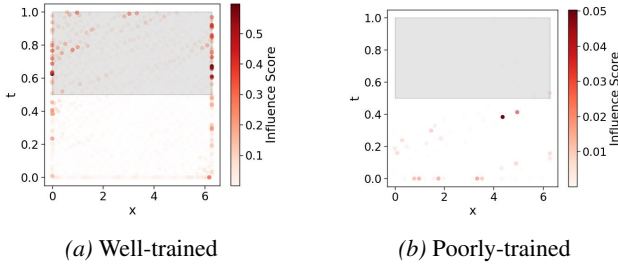


Figure 4. Point-to-region influence heatmaps showcasing the distribution of influences across the training set affecting the network prediction at the top half ($t > 0.5$, marked in gray) of the domain.

points to which these predictions are most sensitive. For the well-trained model, the prediction in that region is primarily shaped by information from the data points on the periodic boundaries ($x = 0$ and $x = 2\pi$). For the poorly-trained model, the influence patterns differ significantly: alongside the region of failure, the prediction is highly sensitive to training points along the initial condition ($t = 0$), while influence from the boundary appears absent. Thus, PINN-FLUENCE reveals a key distinction invisible to residuals: which training data shapes predictions differs drastically between configurations.

Decomposition by Individual Loss Components Point-wise attribution reveals the training points to which the models are most sensitive. Using the linearity of influence functions in the loss argument, PINN-FLUENCE also allows us to directly decompose and quantify influence contributions into individual loss terms (6). This enables studying the contributions of loss terms on a more global scale.

We aggregate the fractional influence contribution $\bar{r}_{L_i}$ of each loss component (the PDE residual, the initial and the two periodic boundary conditions), averaged over time bins in the test data, as shown in Figure 5.

For the well-trained models, the IC contribution starts at around $\bar{r}_{L_{ic}} \approx 0.25$ near $t = 0$ and decreases asymptotically towards zero with increasing time. Simultaneously, the in-

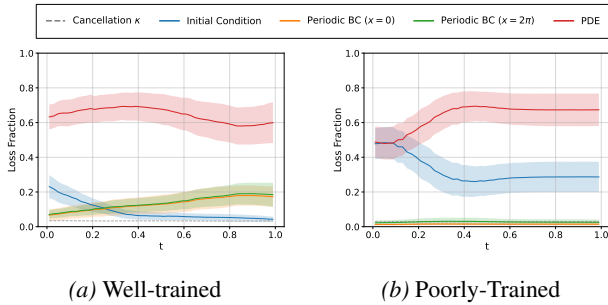


Figure 5. Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{te})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation across seeds.

fluence contributions of both periodic boundary conditions increase evenly over time, indicating that the model’s predictions reflect the problem’s periodic structure. The PDE term dominates throughout.

For the poorly-trained model, the IC displays a much greater share of influence. While it also decreases with time, the term’s relative influence $\bar{r}_{L_{ic}}$ remains elevated in comparison to the well-trained model. After $t \approx 0.4$, the fractions plateau, coinciding with pathological, near-constant outputs and stagnating gradients. Notably, the boundary conditions exhibit negligible contributions, i.e., the model’s predictions are effectively insensitive to its boundary training data. The model has converged to a state where perturbations to boundary points have no effect, a clear structural misalignment for a problem driven by boundary propagation.

We further show that the elevated relative IC importance $\bar{r}_{L_{ic}}$ can be identified across all investigated problems, as depicted in Figure 6. Throughout all well-trained models, the IC influence share at $t = 0$ is lower and decays more rapidly compared to their poorly-trained counterparts. For the latter, the IC generally displays greater influence over time. Note that while the rate of decay varies by problem, the general diagnostic insight holds: well-trained models show diminishing IC sensitivity with time, poorly-trained models do not. This is striking, as not all suboptimally trained models perform equally *poorly*. We refer the interested reader to Appendix B.3, where all respective model predictions and targets are visualized. PINN-FLUENCE can thus attribute model sensitivity to individual loss components L_i , thereby reliably distinguishing well-trained from poorly-trained models.

Indicator-based Analysis The loss decomposition reveals that poorly-trained models are overly sensitive to changes in the initial condition. We now examine this pattern from a complementary perspective: the temporal distribution of influences across the domain. Time-dependent PDEs possess an inherent causal structure in which initial conditions determine future states. One might therefore hypothesize

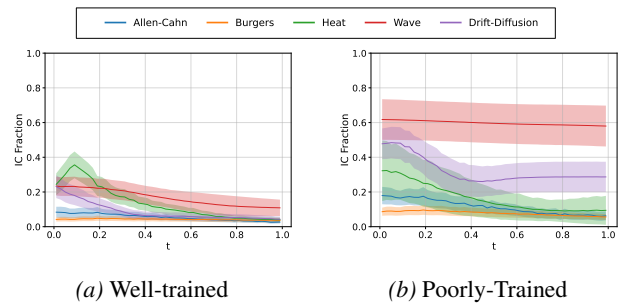


Figure 6. Fractions of IC influences $\bar{r}_{L_{ic}}(\mathcal{X}, R_{te})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation across seeds.

Table 2. Mean temporal influence indicator for each problem, computed with respect to predictions $f = \hat{u}$. We report mean $\pm$ standard deviations over 10 seeds.

Problem	$\bar{t}$	Well-trained	Poorly-trained
Heat	0.46	0.33 ± 0.02	0.26 ± 0.06 (*)
Allen-Cahn	0.43	0.50 ± 0.02	0.32 ± 0.05
Burgers'	0.43	0.41 ± 0.02	0.28 ± 0.02
Drift-Diffusion	0.46	0.46 ± 0.04	0.21 ± 0.06
Wave	0.43	0.41 ± 0.03	0.11 ± 0.02

(*) The baseline is $\bar{t} = 0.41$ for the poorly-trained model due to different sampling.

that a well-trained PINN that yields an accurate approximation of the solution would reflect this causality. Under this proposition, predictions at time t would be predominantly influenced by training points at times $\leq t$. We probe this hypothesis using the temporal influence indicator (9) and compare it against the baseline temporal average $\bar{t} = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} x_t$ of the training set. Note that sampling on the initial boundary introduces a distribution skewed toward earlier times.

Table 2 reports temporal influence indicators across all time-dependent problems, computed with respect to the prediction. Contrary to the aforementioned hypothesis, well-trained models do not exhibit a temporal influence aligned with the causal structure of the underlying PDEs. Rather, they show an influence distribution that is close to the data distribution baseline $\bar{t}$, only deviating slightly from it. This shows that well-trained models are, on average, equally sensitive to changes in training data at both earlier and later times. This is sensible given that PINNs are trained to solve a given problem globally, instead of sequentially. Hence, this illustrates that a well-trained PINN has learned to leverage information from across the spatio-temporal domain. Poorly-trained models, on the other hand, display skewed temporal influences. This finding aligns directly with the elevated IC influence ratio observed earlier: these models show disproportionate sensitivity to early-time training points as exemplified in Figure 6. Hence, strong causal alignment in terms of influences is a signal of training failure rather than of success.

The previous results constitute only a selection of potential questions that one may study using PINNfluence. In Appendix C, we present complementary use-cases, such as additional point-wise attribution heatmaps for other problems, the decomposition of loss terms for each setting. We also introduce further indicators, including a demonstration of region-to-region influence analysis. Notably, we also study steady-state problems with appropriately constructed influence-based indicators.

To summarize, PINNfluence offers a complementary per-

spective on PINNs by attributing predictions directly to the training data and losses. It reliably distinguishes between well-trained and poorly-trained PINNs across diverse problems, enhancing the understanding of PINNs.

Limitations PINNfluence inherits limitations from influence functions. The first-order approximation of leave-one-out retraining is fragile (Basu et al., 2020a) and assumes well-behaved loss landscapes, yet PINNs often exhibit non-convex, ill-conditioned optimization surfaces leading to near-singular Hessians (Rathore et al., 2024). Nevertheless, IFs were found to be accurate at identifying influential training samples, even under large perturbations (Koh et al., 2019; Basu et al., 2020b). While the explicit computation of the inverse Hessian can be avoided through inverse-Hessian vector products (Appendix A.2), it still requires costly approximation. Additionally, the pairwise evaluation of influences scales poorly with training and test set size.

Our methodological choices introduce further caveats: Aggregating absolute influences discards sign information, which is partially addressed by the cancellation score κ . Furthermore, indicator design is problem-dependent and requires domain knowledge for appropriate construction. Interpretation also requires care: influences quantify sensitivity to perturbations in training data, not causal importance. High attribution does not imply necessity or sufficiency for correct behavior. In particular, the temporal indicator should be read as a diagnostic signal rather than a statement about physical causality.

6. Conclusion

Existing studies of PINNs focus primarily on training dynamics and failure modes, but do not provide attribution-based explanations of how training data and physics constraints shape predictions. To address this, we propose PINNfluence, a novel, training-data-centric interpretability method that adapts influence functions to PINNs and their composite losses. By enabling attribution between training data, loss components, and model predictions, PINNfluence supports fine-grained diagnostics of PINN behavior that go beyond residual-based analysis. Our experiments show that loss-component and temporal influence patterns can distinguish well-trained from poorly-trained PINNs, localize failure modes, and reveal structural properties of learned solutions. Overall, while not without limitations, our method presents a principled diagnostic lens for studying PINNs with the aim of imbuing scientific machine learning with interpretability methods. In future work, exploring the use of these methods for intervening in training and informed model design emerges as a promising next step.

Impact Statement

This work introduces a training-data-centric interpretability framework for physics-informed neural networks. By enabling practitioners to attribute model behavior to specific training points and loss components, PINNfluence provides diagnostic capabilities that complement existing methods. We hope this opens up new directions for understanding PINN behavior and, in future work, for developing principled interventions based on this method.

As an interpretability method for scientific machine learning, we do not anticipate negative societal impacts from this work.

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Appendix

The Appendix is structured as follows: [Appendix A](#) focuses on influence functions and provides an extended theoretical background in [Appendix A.1](#) and some additional practical considerations in [Appendix A.2](#).

[Appendix B](#) contains descriptions of the investigated problems, including two steady-state PDEs ([Appendix B.1](#)). [Appendix B.2](#) specifies the training environment and hyperparameters for model configurations across problems and showcases predictions in [Appendix B.3](#).

Finally, [Appendix C](#) features supplemental experiments utilizing PINNfluence. Concretely, we adapt the temporal indicator for steady-state problems in [Appendix C.1](#), and showcase additional region-to-region indicators in [Appendix C.2](#). Furthermore, we also provide loss-term decompositions for the remaining problems [Appendix C.3](#) and additional point-wise heatmaps [Appendix C.4](#).

A. Supplementary Information for Influence Functions

A.1. Theoretical Background

In this part, we formulate the key steps for deriving influence functions, initially introduced in ([Hampel, 1974](#)) to study robust estimators, and which were recently applied to neural networks ([Koh & Liang, 2017](#); [Koh et al., 2019](#); [Hammoudeh & Lowd, 2024](#)). However, as already observed in ([Schioppa et al., 2023](#)), the functions one typically encounters in neural networks require a more exhaustive analysis. Therefore, we carefully redraw the results proposed in ([Koh & Liang, 2017](#)) and generalize them slightly for the application to PINNs. In particular, we formulate influence functions in a manner that enables the study of the effect of various modifications of the problem (i.e., changes in the training set, in the loss weighting, or in the PDE) on the prediction and any derived quantity. Furthermore, we provide rigorous proofs for the statements made in [Section 2.2](#).

We start with a crucial lemma for influence functions, which was already noted in ([Cook & Weisberg, 1982](#)). As remarked in ([Schioppa et al., 2023](#)), influence functions are usually stated for strictly convex functions² in the vicinity of global minima. However, in practice, we typically only reach saddle points ([Dauphin et al., 2014](#)), which is the reason why the following lemma is extended to stationary points.

Lemma A.1 (argmin Trick). *Consider an open set of parameters $\Theta \subseteq \mathbb{R}^p$ and an open set $U \subseteq \mathbb{R}$. Let $g : \Theta \times U \rightarrow \mathbb{R}$ be a twice continuously differentiable function and $(\theta_0, \epsilon_0) \in \Theta \times U$, such that θ_0 is a non-degenerate stationary point of $g(\cdot, \epsilon_0)$, i.e., $\nabla_{\theta} g(\theta_0, \epsilon_0) = 0$ and the Hessian $\mathcal{H}_{\theta}(\theta_0, \epsilon_0) := \nabla_{\theta}^2 g(\theta_0, \epsilon_0) \in \mathbb{R}^{p \times p}$ is invertible.*

Then, there exists a non-empty, open neighborhood $U_0 \subseteq U$ around ϵ_0 , such that $g(\cdot, \epsilon)$ has a unique stationary point in the neighborhood of θ_0 for all $\epsilon \in U_0$. Denote by $h : U_0 \rightarrow \Theta$ the function which describes such stationary points. Then h is continuously differentiable and

$$\frac{\partial h(\epsilon)}{\partial \epsilon} = -\mathcal{H}_{\theta_0}(h(\epsilon), \epsilon)^{-1} \cdot \nabla_{\theta} \frac{\partial g(\theta, \epsilon)}{\partial \epsilon} \Big|_{\theta=h(\epsilon)}. \quad (10)$$

Furthermore, if g is analytic in a neighborhood of (θ_0, ϵ_0) , then we can choose U_0 such that h is analytic as well. Moreover, if $B \subseteq U$ is an open, connected set, such that for every $\epsilon \in B$, there exists a non-degenerate stationary point of $g(\cdot, \epsilon)$, then h is extendable to B and $h : B \rightarrow \Theta$ is also continuously differentiable with (10).

Proof. Let us denote the gradient of g as $s(\theta, \epsilon) := \nabla_{\theta} g(\theta, \epsilon)$. By our initial assumption, g is twice continuously differentiable, implying that s is continuously differentiable. Since (θ_0, ϵ_0) is a stationary point, $s(\theta_0, \epsilon_0) = 0$, the implicit function theorem (see [Oliveira, 2012](#), Theorem 2) guarantees the existence of an open set $U_0 \subseteq U$ around ϵ_0 and a unique, continuously differentiable function $h : U_0 \rightarrow \Theta$ such that $h(\epsilon_0) = \theta_0$ and $s(h(\epsilon), \epsilon) = 0$ for all $\epsilon \in U_0$. This establishes that for any $\epsilon \in U_0$, a unique stationary point exists in the neighborhood of θ_0 . And h is continuously differentiable with (10).

Now if g is analytic, then so is its gradient s . Therefore also the function h is analytic, following the holomorphic implicit function theorem (see [Fritzsche & Grauert, 2002](#), Section 7.6).

²Note, that for the application of influence functions with global minima (e.g. [Koh & Liang, 2017](#)), strict convexity is a necessary, but not a sufficient criterion.

If for all $\epsilon \in B$ the above assumptions are fulfilled, then we can apply [Lemma A.1](#) for all of these points ϵ . The function $h : B \rightarrow \Theta$ is continuous differentiable, since it is continuous differentiable in every point and (10) extends to all points in B . Moreover, $h(B)$ is connected, since h is continuous and B is connected by assumption. $\square$

By means of this theorem, we can now prove [Theorem 2.2](#) from [Section 2.2](#). For better readability, we restate that theorem.

Theorem A.2 (Influence Functions Approximate Retraining, analogue to [Theorem 2.2](#)). *Assume the setting of [Definition 2.1](#) with L and f being analytic, and that θ_0 is a non-degenerate stationary point of $\mathcal{L}$ with respect to a training set $\mathcal{X}$. Let θ_1 denote the stationary point of $\mathcal{L}(\theta)$ (i.e., $\nabla_{\theta}\mathcal{L}(\theta_0) = 0$, $\det \nabla_{\theta}^2\mathcal{L}(\theta_0) \neq 0$) with respect to $\mathcal{X}'$, where we added or removed a training point $x^{\pm}$ (with $x^- \in \mathcal{X}$). For sufficiently large N , the resulting effect on the function f is given by*

$$f(z, \theta_1) - f(z, \theta_0) = \pm \text{Inf}_{\theta_0}^{L \rightarrow f}(x^{\pm}, z) \frac{1}{N} + \mathcal{O}(1/N^2) \quad (11)$$

Proof. Let $\tilde{\mathcal{L}}(\theta, \epsilon) := \mathcal{L}(\theta) + \epsilon \kappa(\theta)$ be a small perturbation of the original loss with some differentiable function $\kappa : \Theta \rightarrow \mathbb{R}$ and $\epsilon \in \mathbb{R}$. Since θ_0 is a non-degenerate stationary point of $\mathcal{L}(\theta)$, $(\theta_0, 0)$ is a non-degenerate stationary point of $\tilde{\mathcal{L}}(\theta, \epsilon)$ in the sense of [Lemma A.1](#). Hence, there is a neighborhood U_0 around the origin $\epsilon_0 = 0$, such that the stationary points θ_{ϵ} of $\tilde{\mathcal{L}}(\theta, \epsilon)$ are uniquely defined for all $\epsilon \in U_0$ and the derivative of θ_{ϵ} is given by (10).

Since θ_{ϵ} is also analytic, due to the assumptions and [Lemma A.1](#), we can approximate it with a Taylor expansion as

$$\theta_{\epsilon} - \theta_0 = \frac{\partial \theta_{\epsilon}}{\partial \epsilon} \Big|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2) = -\mathcal{H}_{\theta_0}^{-1} \cdot \nabla_{\theta} \kappa(\theta) \Big|_{\theta=\theta_0} \cdot \epsilon + \mathcal{O}(\epsilon^2) \quad (12)$$

with $\mathcal{H}_{\theta_0} = \nabla_{\theta}^2 \mathcal{L}(\theta) \Big|_{\theta=\theta_0}$. When setting $\kappa(\theta) = L(x^+; \theta)$, we have $\tilde{\mathcal{L}}(\theta, 1/N) = \frac{1}{N} \sum_{x \in \mathcal{X} \cup \{x^+\}} L(x; \theta)$, which has the same stationary points as the empirical risk over $\mathcal{X} \cup \{x^+\}$, which is $\frac{1}{N+1} \sum_{x \in \mathcal{X} \cup \{x^+\}} L(x; \theta)$. In other words, we have $\theta_{1/N} = \theta_1$ providing that $1/N \in U_0$, which is always satisfied for large N . We obtain by chain rule

$$f(z, \theta_1) - f(z, \theta_0) = \nabla_{\theta} f(z, \theta_0) \cdot \frac{\partial \theta_{\epsilon}}{\partial \epsilon} \Big|_{\epsilon=0} \cdot \frac{1}{N} + \mathcal{O}(1/N^2) \quad (13)$$

which concludes the proof for x^+ by application of [Lemma A.1](#). Similarly, we obtain the result for x^- by considering $\kappa(\theta) = -L(x^-; \theta)$ with an additional sign.

The [Corollary 2.3](#) follows immediately, by considering the linearity in equations (12) and (13). $\square$

From the details of the proof we also immediately see the effect of adding several or removing several points. Hence, we find that a change from a training set $\mathcal{X}$ to a set $\mathcal{X}' = \mathcal{X} \cup \mathcal{X}^+ \setminus \mathcal{X}^-$, where we assume $\mathcal{X}^- \subset \mathcal{X}$ will affect a function $f : A \times \Theta$ according to

$$f(z; \theta_1) - f(z; \theta_0) = \frac{1}{N} \left(\sum_{x \in \mathcal{X}^+} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) - \sum_{x \in \mathcal{X}^-} \text{Inf}_{\theta_0}^{L \rightarrow f}(x, z) \right) + \mathcal{O}(1/N^2) \quad (14)$$

for large $N > N_{\min}$. There is always a value $N_{\min}$, where (14) is valid, but the precise number depends on the set U_0 , and for large changes of the training set $\mathcal{X} \rightarrow \mathcal{X}'$ this number might increase.

One can easily extend [Theorem A.2](#) to study also the influence of other parts in the optimization task. E.g. one could study the influence of certain parameters in the PDE, such as the diffusivity constant α in the heat equation. By computing the Taylor expansion with respect to that parameter α , one approximate the effect of increasing or lowering that parameter. The proof is completely analogue to the proof of [Theorem A.2](#) and is left to the reader.

A.2. Practical Considerations On The Computation Of Influences

Note that the explicit computation of the inverse Hessian $\mathcal{H}_{\theta}^{-1}$ is extremely costly, however we can circumvent this by utilizing inverse-Hessian vector products (IHVPs). Several efficient approximations of IHVPs have already been proposed in the literature, such as linear time stochastic second-order algorithm (LiSSA) ([Agarwal et al., 2017](#)), Arnoldi iterations

(Schioppa et al., 2021) and Eigenvalue-corrected Kronecker-Factored Approximate Curvature (EK-FAC) (Grosse et al., 2023). In this study we utilize a modified version of the `ArnoldiInfluenceFunction` from the `Captum` library (Kokhlikyan et al., 2020).

The first-order approximation underlying influence functions assumes convexity and is found to be fragile as a true leave-one-out estimator (Basu et al., 2020a), especially for very deep neural networks. While our model configurations are fairly shallow, especially in comparison to modern computer vision and LLM architectures, PINNs exhibit non-convex and ill-conditioned loss landscapes (Rathore et al., 2024). Nevertheless, despite these theoretical limitations influence functions have been found to successfully recover influential samples in practice (Koh et al., 2019; Basu et al., 2020b). Recent work suggests that they are better understood as approximators of local fine-tuning behavior rather than exact leave-one-out-retraining (Bae et al., 2022; Wei et al., 2024).

Beyond influence functions, alternative training data attribution methods exist with different trade-offs. Retraining-based approaches such as naive leave-one-out retraining and Data Shapley (Ghorbani & Zou, 2019) provide gold standard estimates but are computationally prohibitive. Simpler approximations include similarity-based methods (Charpiat et al., 2019) that compare loss gradients or penultimate-layer representations (Pezeshekpour et al., 2021). More sophisticated alternatives include the Representer Points method (Yeh et al., 2018), which avoids Hessian computation via L^2 regularization and application of the representer theorem. TracIn (Pruthi et al., 2020) accumulates gradient dot products throughout training to trace influence over the optimization trajectory. DualXDA (Yolcu et al., 2025) performs attribution in the penultimate layer using SVM dual coefficients for sparse and efficient estimates, further employing relevance back-propagation (Bach et al., 2015) to pinpoint influential features in a given training sample.

B. Problem Specifications

B.1. Investigated Partial Differential Equations

Allen-Cahn Equation: A nonlinear reaction-diffusion equation modeling phase separation in materials which is given by

$$\frac{\partial u}{\partial t} - \epsilon^2 \frac{\partial^2 u}{\partial x^2} + f(u) = 0 \quad . \quad (15)$$

We take $\Omega = (-1, 1) \times (0, 1)$, select $f(u) = -5(u^3 - u)$ and $\epsilon = \sqrt{10^{-4}}$, with the following boundary conditions $u(x, 0) = x^2 \cos(\pi x)$, $u(-1, t) = -1$ and $u(1, t) = -1$.

In the specified setting the reaction term $-5(u^3 - u) = -5u(u^2 - 1)$ has two stable wells at $u = \pm 1$ for regions with negligible diffusion. The setup drives negative u to -1 and positive to 1 , respectively. Note however, that the reaction term may also collapse to 0 for $u = 0$, allowing for low PDE residuals with an incorrect solution. We obtained reference values from (Wu et al., 2023) shown in Figure 7.

Burgers' Equation: A second-order non-linear PDE that models phenomena combining convection and diffusion, often used as a simplified model for fluid dynamics or traffic flow. It is given by

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0 \quad . \quad (16)$$

We take $\Omega = (-1, 1) \times (0, 1)$ and set the viscosity to $\nu = \frac{10^{-3}}{\pi}$, with the following boundary conditions $u(x, 0) = -\sin(\pi x)$, $u(-1, t) = 0$ and $u(1, t) = 0$. In this setup it develops a sharp shock around $x = 0$. We obtained reference values from (Wu et al., 2023) shown in Figure 8.

Drift-Diffusion Equation: A second order linear PDE describing diffusion with coefficient α with given drift β

$$\frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} + \beta \frac{\partial u}{\partial x} = 0 \quad (17)$$

We take $\Omega = (0, 2\pi) \times (0, 1)$ and set $\alpha = 1, \beta = 20$, with the following initial and boundary conditions $u(x, 0) = \sin(2x + \frac{\pi}{4})$, $u(0, t) = \sin(\frac{\pi}{4} - 2 \cdot \beta t) e^{-4\alpha t}$ and $u(2\pi, t) = \sin(\frac{17\pi}{4} - 2\beta t) e^{-4\alpha t}$. This yields the closed form solution

$$u(x, t) = \sin\left(2x - 2\beta t + \frac{\pi}{4}\right) e^{-4\alpha t} \quad . \quad (18)$$

A similar problem was found to cause propagation failures in PINNs (Krishnapriyan et al., 2021), where the PINN fails to generalize over the domain, which motivated the inclusion of this example as a prominent problem in the main text. We provide a visualization in Figure 9.

Heat Equation: A second-order linear PDE for modeling diffusion

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - s(x) = 0 \quad , \quad (19)$$

where $s : \Omega \rightarrow \mathbb{R}$ denotes a source term, which we set to $s(x, t) := (\pi^2 - 1) \sin(\pi x) e^{-t}$.

We chose $\Omega = (-1, 1) \times (0, 1)$, with the following boundary conditions $u(x, 0) = \sin(\pi x)$, $u(-1, t) = 0$ and $u(1, t) = 0$. Which yields the closed form solution (Wu et al., 2023).

$$u(x, t) = \sin(\pi x) e^{-t} \quad (20)$$

The heat equation is a considerably simple PDE for basic benchmarks. See Figure 10 for visualization.

Wave Equation: A second-order linear PDE describing the propagation of waves (e.g., vibrations, acoustics), given by

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad (21)$$

We take $\Omega = (0, 1) \times (0, 1)$, set $c = 2$, with the following boundary conditions $u(x, 0) = \sin(\pi x) + \frac{1}{2} \sin(4\pi x)$, $\frac{\partial u}{\partial t}(x, 0) = 0$, $u(0, t) = 0$ and $u(1, t) = 0$. Which yields the following closed form solution

$$u(x, t) = \sin(\pi x) \cos(\pi c t) + \frac{1}{2} \sin(4\pi x) \cos(4\pi c t). \quad (22)$$

This IBVP yields antisymmetric patterns as well as high and low frequency changes in the solution shown in Figure 11.

Navier-Stokes Equations: A system of second-order non-linear PDEs describing the motion of viscous fluids, such as water or air. We follow the formulation from the DFG flow around cylinder benchmark in the laminar case with the Reynolds number being set to 20 (Schäfer et al., 1996). This results in an incompressible and steady instance of the Navier-Stokes equations, which are given in this case as

- *x*-Velocity: $u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{1}{\rho} \frac{\partial p}{\partial x} - \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0$
- *y*-Velocity: $u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{1}{\rho} \frac{\partial p}{\partial y} - \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) = 0$
- Continuity: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

where $(u, v) \in \mathbb{R}^2$ denotes the velocity field, $p \in \mathbb{R}$ the pressure, ρ the fluid density and ν the kinematic viscosity. A Reynolds number of 20 induces $\rho = 1$ and $\nu = \frac{1}{1000}$. We solve the non-dimensionalized form of these equations, scaling coordinates by characteristic length $L^* = 0.1$ and velocities by $U^* = 0.2$. The domain is set to $\Omega = (0, 2.2) \times (0, 0.41) \setminus C_r(0.2, 0.2)$, where C_r denotes a cylinder with center at $(0.2, 0.2)$ and radius $r = 0.05$. The boundary conditions are given as follows, the inflow is characterized by a parabola $u(0, y) = \left(\frac{4Uy(0.41-y)}{0.41^2}, 0 \right)$ where $U = 0.3$ denotes the maximum inflow velocity. As boundary conditions for the lower and upper walls $\Gamma_{\text{bottom}} = (0, 2.2) \times \{0\}$ and $\Gamma_{\text{top}} = (0, 2.2) \times \{0.41\}$ as well as on the cylinder boundary $\Gamma_{\text{cylinder}} = \partial C_r(0.2, 0.2)$, the predicted *x* and *y*-velocity should be 0.

Furthermore, for the outflow on the right edge, i.e., $\Gamma_{\text{outflow}} = \{2.2\} \times (0, 0.41)$ do-nothing boundary conditions are defined, i.e., $\nu \frac{\partial u}{\partial x} - p$ and $\nu \frac{\partial v}{\partial y}$ should be 0.

Target values are obtained from (Wang et al., 2023) and are visualized in Figure 12.

Poisson Equation with a point source over a unit disk: The Poisson equation models the electrostatic potential induced by an electric point charge and is given by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \delta_\sigma(x) = 0 \quad , \quad (23)$$

where δ_σ denotes a Gaussian approximation of the Dirac delta with bandwidth $\sigma = 0.02$. The domain we consider is the unit disk $\Omega = \{x \in \mathbb{R}^2 : \|x\|_2 < 1\}$ with a unit point source at the origin $c = (0, 0)$. The analytical solution corresponding to the Dirac source (recovered in the limit $\sigma \rightarrow 0$) is radially symmetric and is given as

$$u(r) = -\frac{1}{2\pi} \log r \quad , \quad (24)$$

where $r = \|x\|_2$ with $0 < r \leq 1$. We specify the following boundary condition $u(x) = 0$ for $\|x\| = 1$.

This problem yields a difficult to learn singularity and exhibits radial symmetries, which result in an interesting benchmark problem. We provide a visualization in Figure 13.

B.2. Training Setup

All models have been trained with the DeepXDE library (Lu et al., 2021), including the geometry generation and sampling, with PyTorch (Paszke et al., 2019) used as its backend. Optimization was carried out in two stages, first with the Adam optimizer (Kingma & Ba, 2017) followed by fine-tuning with the second order L-BFGS optimizer (Liu & Nocedal, 1989). Note that to construct some of the poorly-trained models, we deliberately skipped the L-BFGS optimization step.

The hyperparameters for each configuration is presented in Table 3. All models were trained as simple feed-forward neural networks with equal number of neurons per hidden layer and tanh activations.

Table 3. Hyperparameters for well-trained and poorly-trained configurations. Layers and neurons specify depth and width of the feed-forward neural networks; N_{domain} , N_{bc} , N_{ic} specify numbers of sampled points inside the domain, on boundaries and on the initial time slice; $N_{\text{iter}}^{(\text{Adam})}$ and $N_{\text{iter}}^{(\text{L-BFGS})}$ specify the number of optimization steps using the respective optimizer.

Problem	Config	Layers	Neurons	N_{domain}	N_{bc}	N_{ic}	$N_{\text{iter}}^{(\text{Adam})}$	$N_{\text{iter}}^{(\text{L-BFGS})}$
Allen-Cahn	Well-trained	3	64	2 500	500	500	100 000	25 000
	Poorly-trained	3	64	2 500	500	500	100 000	—
Burgers'	Well-trained	3	32	2 500	500	500	50 000	12 000
	Poorly-trained	3	32	500	100	100	50 000	12 000
Drift-Diffusion	Well-trained	3	64	1 000	100	100	15 000	5 000
	Poorly-trained	3	64	200	20	20	15 000	5 000
Heat	Well-trained	3	32	1 000	100	100	15 000	5 000
	Poorly-trained	3	32	10	2	2	15 000	5 000
Wave	Well-trained	5	100	2 500	500	500	100 000	25 000
	Poorly-trained	5	100	2 500	500	500	100 000	—
Navier-Stokes	Well-trained	3	64	7 500	2 500	—	100 000	25 000
	Poorly-trained	3	64	1 500	500	—	100 000	25 000
Poisson	Well-trained	3	32	2 500	500	—	50 000	12 000
	Poorly-trained	3	32	100	20	—	15 000	5 000

B.3. Model Predictions and Targets

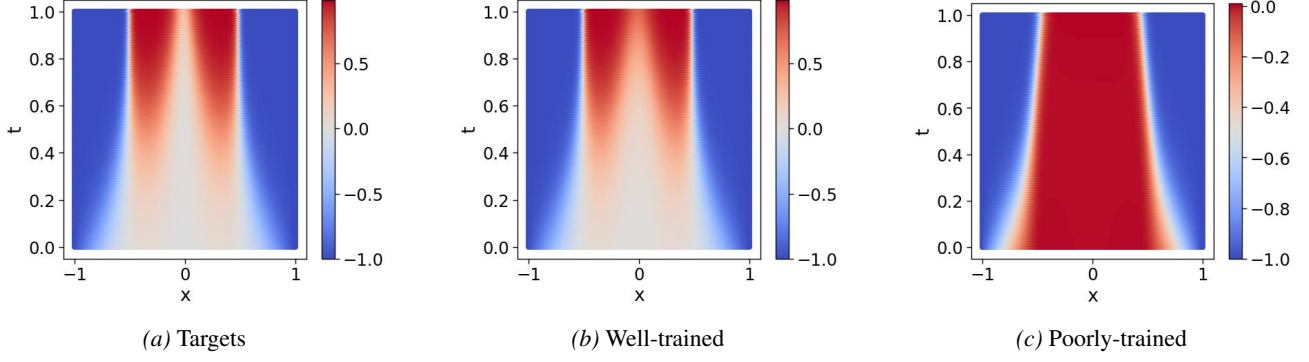


Figure 7. Allen-Cahn equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

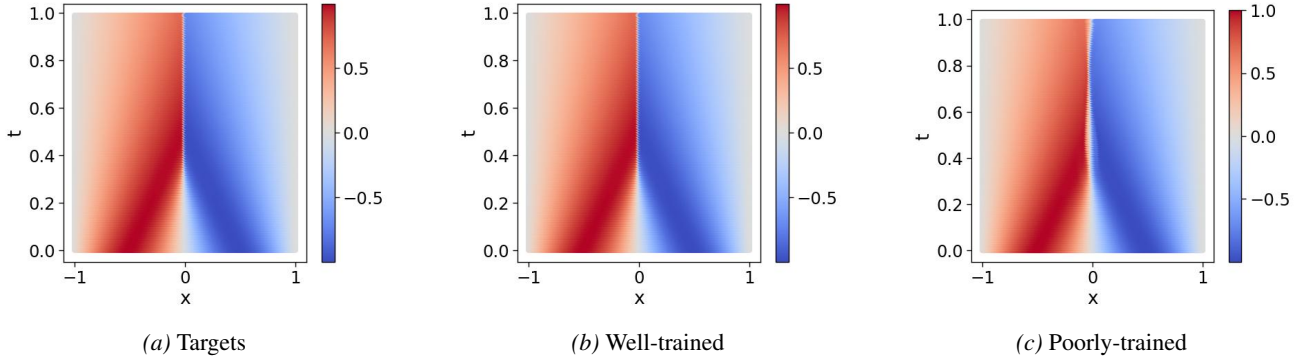


Figure 8. Burgers' equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

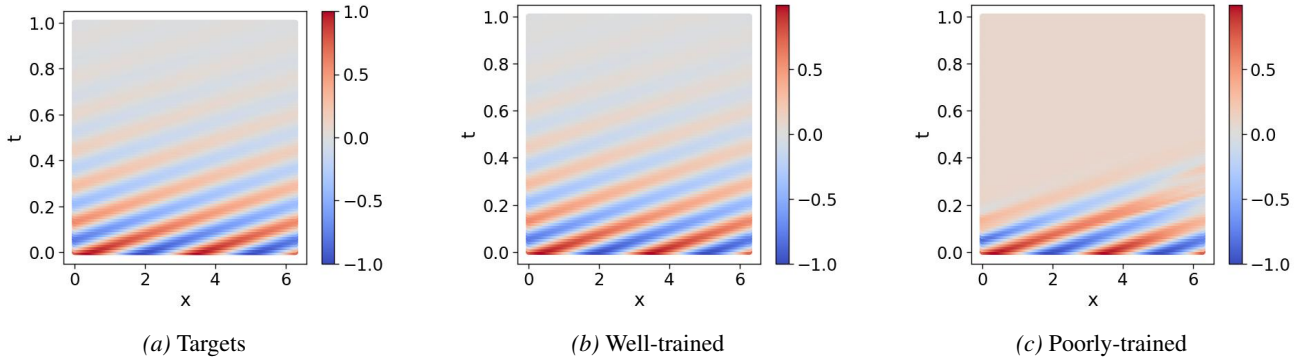


Figure 9. Drift-diffusion equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

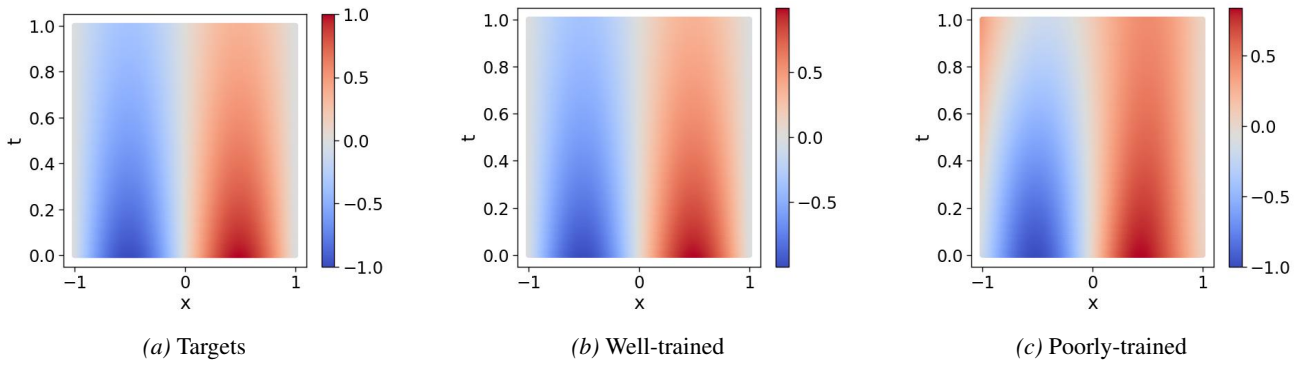


Figure 10. Heat equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

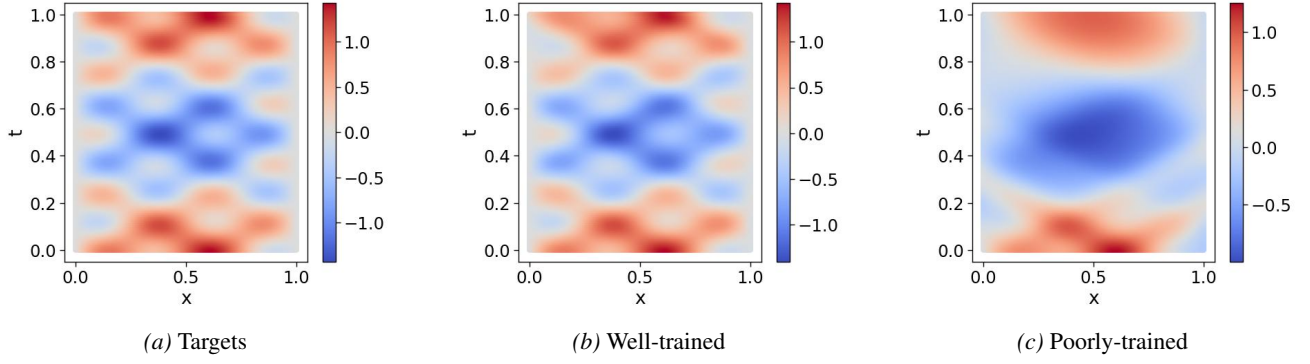


Figure 11. Wave equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

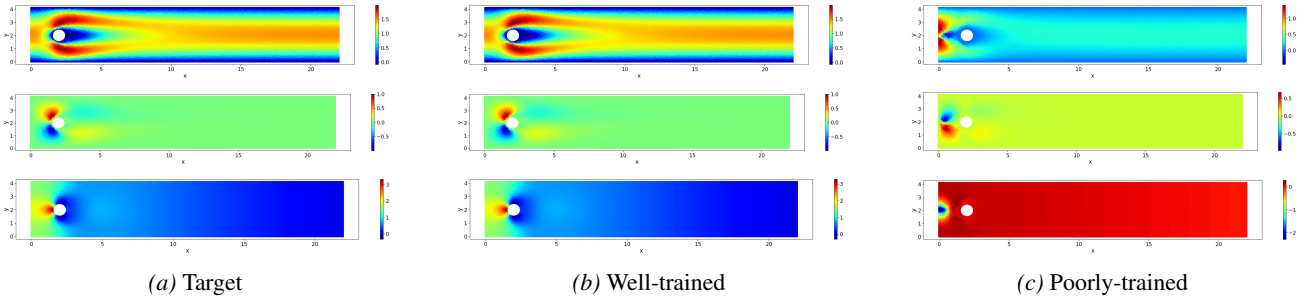


Figure 12. Navier-Stokes equations: target values and predictions for x -velocity u (top row), y -velocity v (middle row), and pressure p (bottom row). Predictions averaged over 10 seeds.

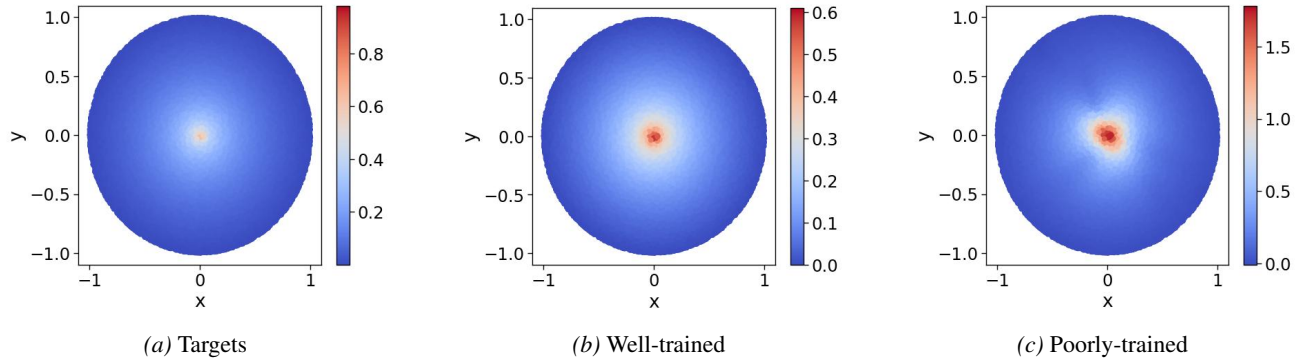


Figure 13. Poisson equation: target values and predictions for well and poorly-trained configurations averaged over 10 seeds.

C. Additional Investigations

C.1. Reframing the Temporal Indicator for Steady-State Problems as a Measure of Directionality

The causality indicator can be extended to the featured steady-state problems, replacing temporal influence with directional influence, appropriate to the problem structure.

For the Navier-Stokes equation, we consider the inflow boundary condition analogous to an initial condition. Here, the flow propagates along the x -axis instead. This results in the analogous indicator to (9) with a different spatial coordinate

$$\eta_{\theta_0}^{L \rightarrow f}(R_{\text{tr}}, R_{\text{te}}) = 1 - \frac{1}{|R_{\text{te}}|} \sum_{z \in R_{\text{te}}} \frac{\sum_{x \in R_{\text{tr}}: x_x \leq z_x} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}{\sum_{x \in \mathcal{X}} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}. \quad (25)$$

For the Poisson equation, we consider the problem stated at a disk domain, featuring a singularity in the center acting as a source. The directional indicator measures influence from training points closer to the center, relative to the total influence

$$\eta_{\theta_0}^{L \rightarrow f}(R_{\text{tr}}, R_{\text{te}}) = 1 - \frac{1}{|R_{\text{te}}|} \sum_{z \in R_{\text{te}}} \frac{\sum_{x \in R_{\text{tr}}: \|x\| \leq \|z\|} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}{\sum_{x \in \mathcal{X}} |\text{Inf}_{\theta_0}^{L \rightarrow f}(x, z)|}. \quad (26)$$

Table 4. Spatial directionality indicator for steady-state problems, evaluated for predictions $f = \hat{u}$. We report mean $\pm$ standard deviation over 10 seeds.

Problem	f	$\bar{x}$	Well-trained	Poorly-trained
Poisson	$\hat{u}$	0.28	0.29 ± 0.03	0.69 ± 0.06
Navier-Stokes	x -velocity	0.48	0.15 ± 0.01	0.15 ± 0.03
	y -velocity	0.48	0.25 ± 0.01	0.2 ± 0.05
	Pressure	0.48	0.14 ± 0.02	0.11 ± 0.02

Table 4 reports directionality indicators. For the well-trained Poisson model, the indicator matches the expected baseline. For the poorly-trained model, the indicator is far above the baseline. Contrary to the time-dependent problems, the model is heavily influenced by points near the boundary. This represents the inverse of the time-dependent pathology: rather than over-relying on the source term the poorly-trained model completely neglects it. Note that it does not feature a separate loss term and is instead modeled solely through the PDE residual.

For the Navier-Stokes problem, we consider steady flow around an obstacle close to the inflow. We find that the directionality indicator shows no separation between the model configurations. This is because influences are concentrated around the obstacle, which is the defining structure of the flow, rather than distributed along the flow direction. The measured indicator values confirm this, as they closely match the ratio of training points downstream versus upstream of the obstacle.

C.2. Additional Indicators

In this section we demonstrate how one may construct problem-specific indicators using PINNfluence to specifically probe individual model characteristics.

Navier-Stokes: Downstream Influence on Individual Regions The region-based influence indicator $\rho_{\theta_0}^{L \rightarrow f}$ (8) allows to quantify the influence of one region in the training set R_{tr} onto another region in the test set R_{test} . We will exemplify this on the Navier-Stokes problem.

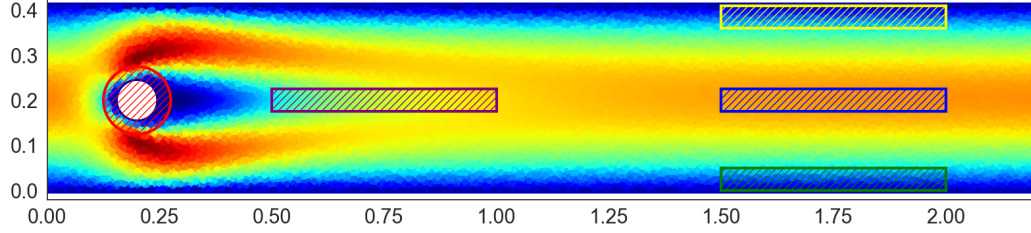


Figure 14. Regions of interest to quantify the influence of the region surrounding the cylinder on downstream locations: (i) closely behind the cylinder; (ii) close to the outflow, centered; (iii) close to the outflow, top; (iv) close to the outflow, bottom.

Table 5. Influence scores from cylinder region for different loss terms and spatial regions. We report mean $\pm$ standard deviation over 10 seeds.

f	Configuration	Center-Rectangle	Outflow-Rectangle	Top-Rectangle	Bottom-Rectangle
x -Velocity	Well-trained	0.43 ± 0.05	0.38 ± 0.05	0.33 ± 0.05	0.36 ± 0.05
	Poorly-trained	0.31 ± 0.09	0.27 ± 0.09	0.26 ± 0.09	0.25 ± 0.09
y -Velocity	Well-trained	0.27 ± 0.04	0.25 ± 0.03	0.22 ± 0.03	0.23 ± 0.02
	Poorly-trained	0.24 ± 0.08	0.23 ± 0.08	0.23 ± 0.09	0.21 ± 0.09
Pressure	Well-trained	0.48 ± 0.07	0.47 ± 0.07	0.47 ± 0.07	0.46 ± 0.07
	Poorly-trained	0.25 ± 0.06	0.25 ± 0.06	0.24 ± 0.06	0.26 ± 0.06
Loss	Well-trained	0.32 ± 0.03	0.28 ± 0.04	0.24 ± 0.03	0.25 ± 0.02
	Poorly-trained	0.26 ± 0.08	0.23 ± 0.08	0.23 ± 0.09	0.22 ± 0.06

The results presented in Table 5 confirm physical expectations: excluding the predictions for pressure, the influence of the cylinder is strongest in the rectangle directly downstream and diminishes toward the outflow. Top and bottom regions show similar values, consistent with the flow’s approximate symmetry. Note, though, that the cylinder is not perfectly centered, hence slight asymmetries are expected. Well-trained models exhibit consistently higher influences than poorly-trained ones, most strikingly for pressure, and lower variance across seeds.

Wave Equation: Symmetry Indicator We test for symmetry in influences for the wave equation by comparing the influence of the left spatial half ($x < 0.5$) onto the right ($x \geq 0.5$) and vice versa.

Table 6. Spatial symmetry indicators for the Wave equation. Values represent the fraction of influence from one spatial half to the opposite half. The ratio quantifies symmetry (values near 1.0 indicate symmetric influence patterns). We report mean $\pm$ standard deviation over 10 seeds.

f	Configuration	$\rho_{\theta_0}^{L \rightarrow f}(R_{\text{left}}, R_{\text{right}})$	$\rho_{\theta_0}^{L \rightarrow f}(R_{\text{right}}, R_{\text{left}})$	$\frac{\rho_{\theta_0}^{L \rightarrow f}(R_{\text{right}}, R_{\text{left}})}{\rho_{\theta_0}^{L \rightarrow f}(R_{\text{left}}, R_{\text{right}})}$
$\hat{u}$	Well-trained	0.48 ± 0.04	0.49 ± 0.05	1.01 ± 0.17
	Poorly-trained	0.48 ± 0.04	0.51 ± 0.04	0.96 ± 0.14
Loss	Well-trained	0.35 ± 0.03	0.36 ± 0.04	0.98 ± 0.18
	Poorly-trained	0.38 ± 0.02	0.40 ± 0.02	0.95 ± 0.08

As shown in Table 6, both configurations display symmetric influence patterns with ratios close to 1. Notably, the influence fractions of 0.48 to 0.51 indicate that predictions in each half are nearly equally shaped by training points from the opposite half, consistent with perfect symmetry. Although the poorly-trained model fails to capture high-frequency features in the solution, these results suggest that it still accurately learns the problem’s spatial symmetry. The loss decomposition, however, clearly separates well-trained from poorly-trained models, see Appendix C.3.

Burgers’ Equation: Locality of the Shock Burgers’ equation develops a sharp shock around $x = 0$. We test how locally distributed influences are around the shock region by computing the region-to-region influence from training points within $x \in [-0.2, 0.2]$ onto test points within the same region.

Table 7. Shock locality indicator for Burgers’ equation. Values represent the fraction of influence on shock region predictions (x coordinates in $[-0.2, 0.2]$) originating from training points within the same region. We report mean $\pm$ standard deviation over 10 seeds.

f	Configuration	$\rho_{\theta_0}^{L \rightarrow f}(R_{\text{shock}}, R_{\text{shock}})$
$\hat{u}$	Well-trained	0.66 ± 0.03
	Poorly-trained	0.38 ± 0.06
Loss	Well-trained	0.71 ± 0.02
	Poorly-trained	0.47 ± 0.05

Table 7 indicates that well-trained models show strong locality, while poorly-trained models exhibit weaker locality, indicating over-sensitivity to regions irrelevant to the shock.

C.3. Decompositions by Loss-Components for other Problems

Here we present the loss decompositions for all problems not featured in the main text.

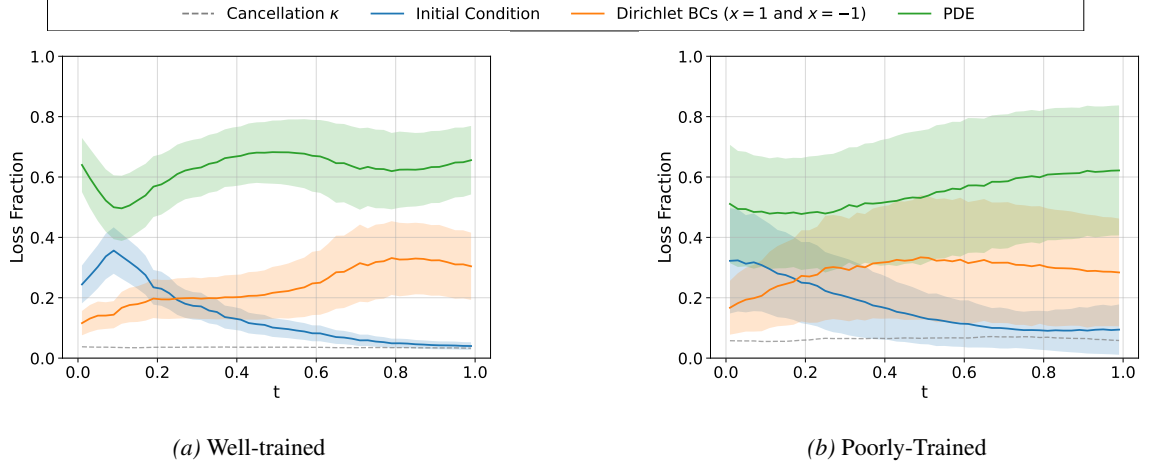


Figure 15. Heat equation: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation.

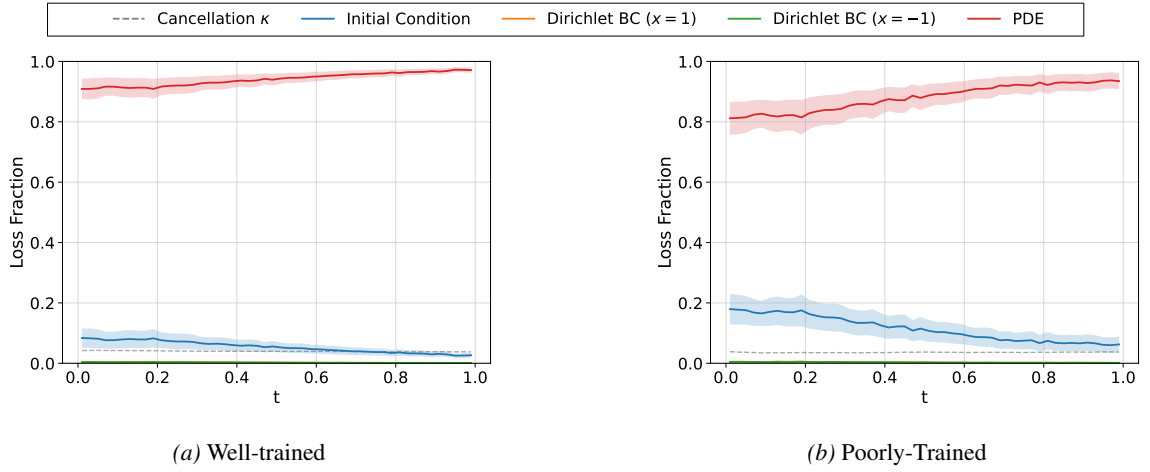


Figure 16. Allen-Cahn equation: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation.

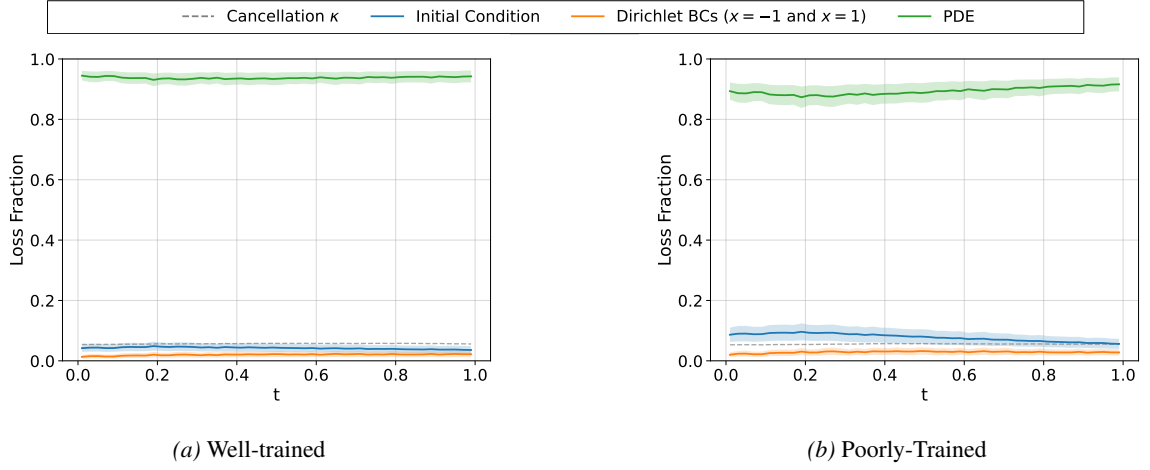


Figure 17. Burgers' equation: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation.

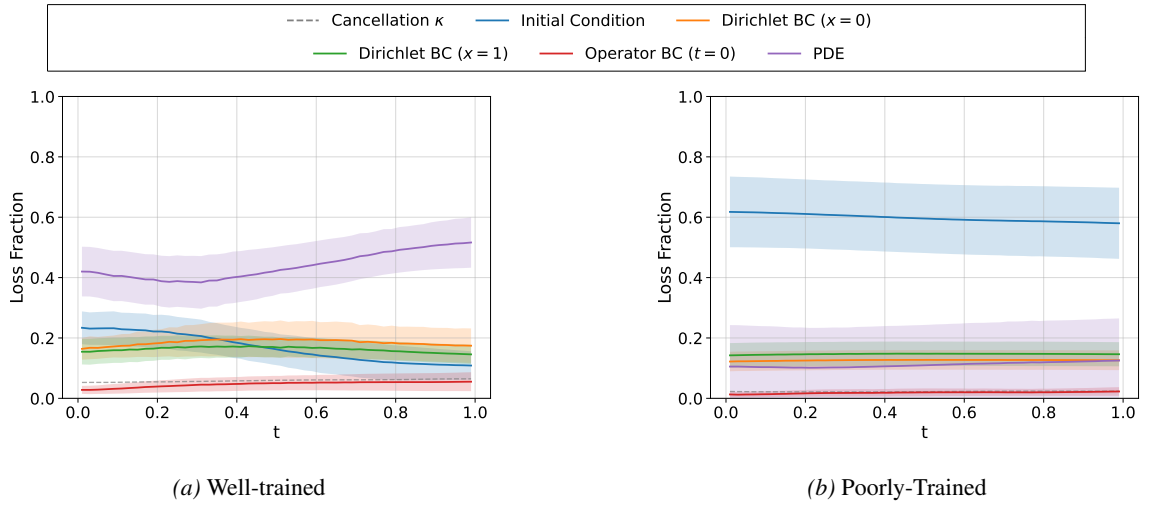


Figure 18. Wave equation: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation.

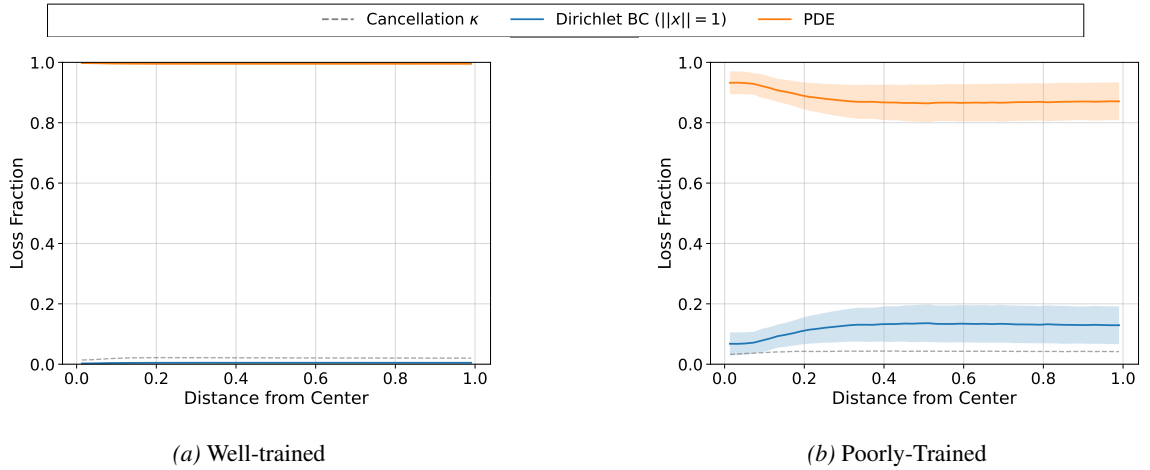


Figure 19. Poisson equation: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation.

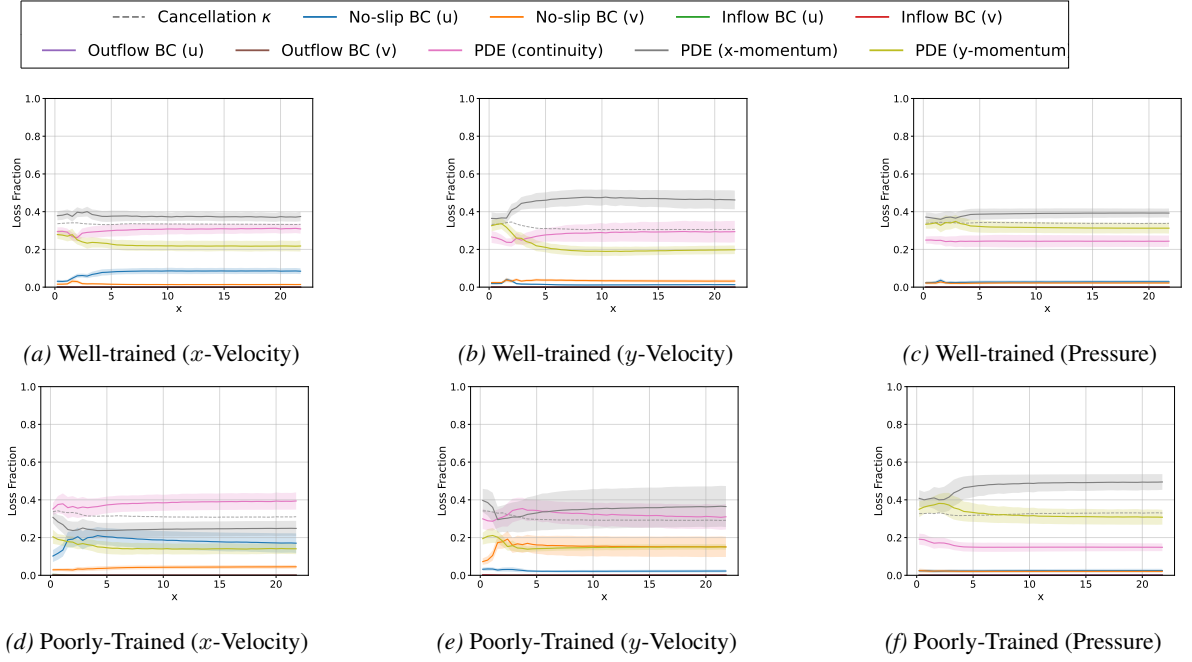


Figure 20. Navier-Stokes equations: Fractions of influences $\bar{r}_{L_i}(\mathcal{X}, R_{\text{test}})$ for 50 time bins, averaged over seeds. Shaded regions indicate the standard deviation. Evaluated for all three outputs respectively.

C.4. Supplementary Point-wise Influence Plots

We visualize point-wise influence scores for each benchmark problem. For two representative points per problem, we compare well-trained and poorly-trained models side by side. Red heatmaps show which training points influence a fixed test point, whereas blue heatmaps show how a fixed training point influences the whole domain.

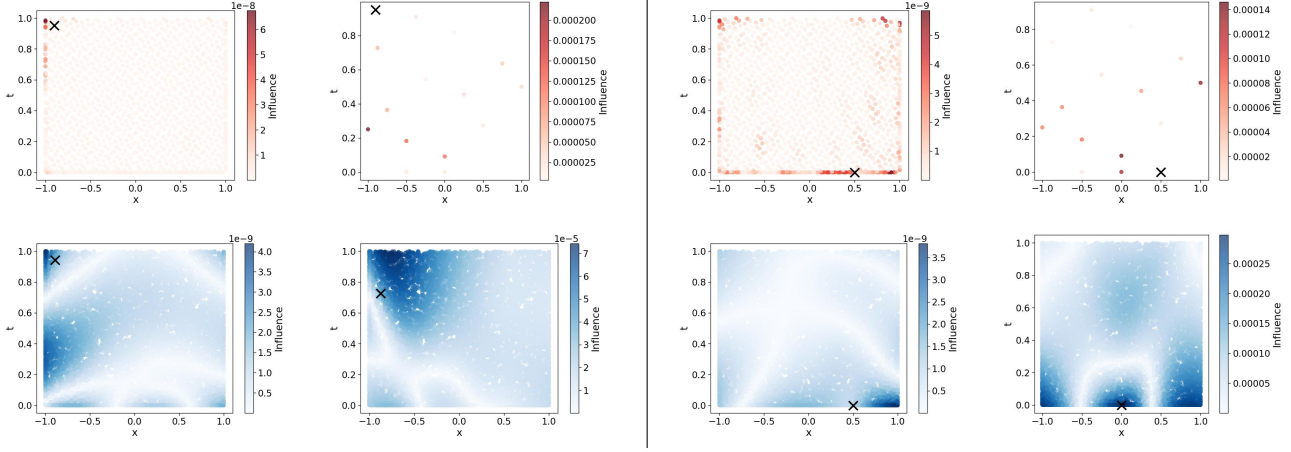


Figure 21. Heat equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

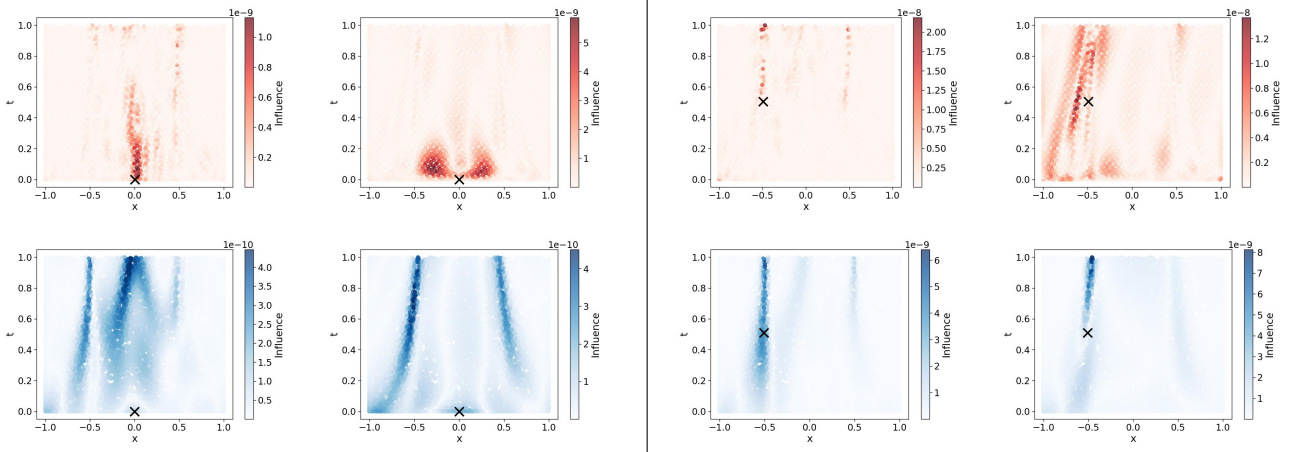


Figure 22. Allen-Cahn equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

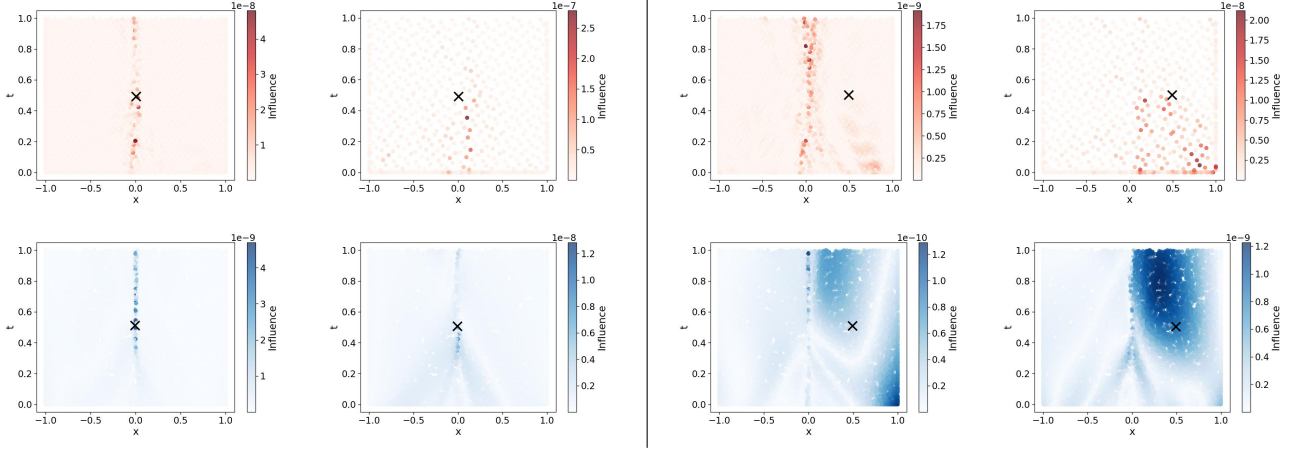


Figure 23. Burgers equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

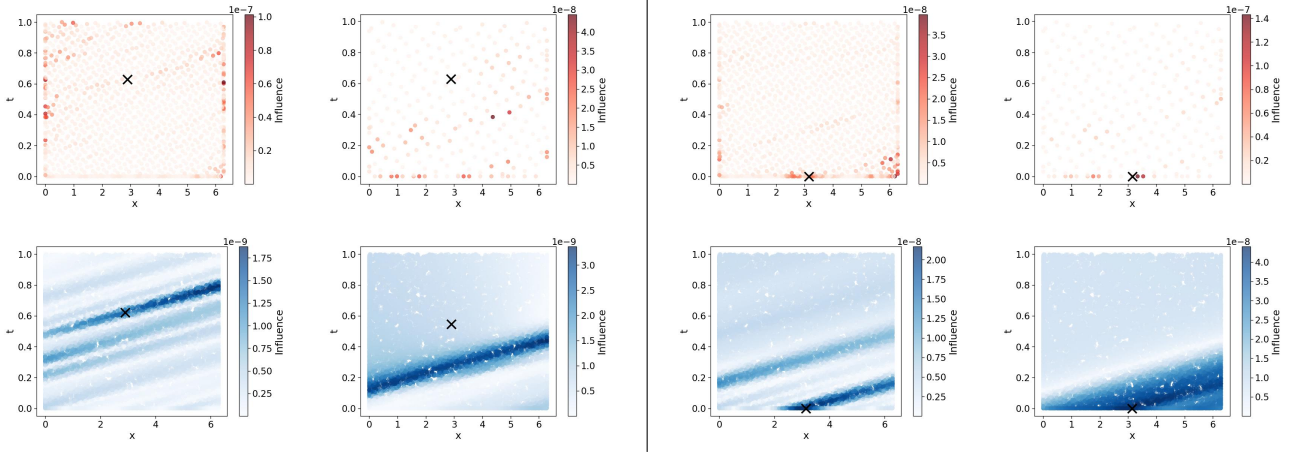


Figure 24. Drift-diffusion equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

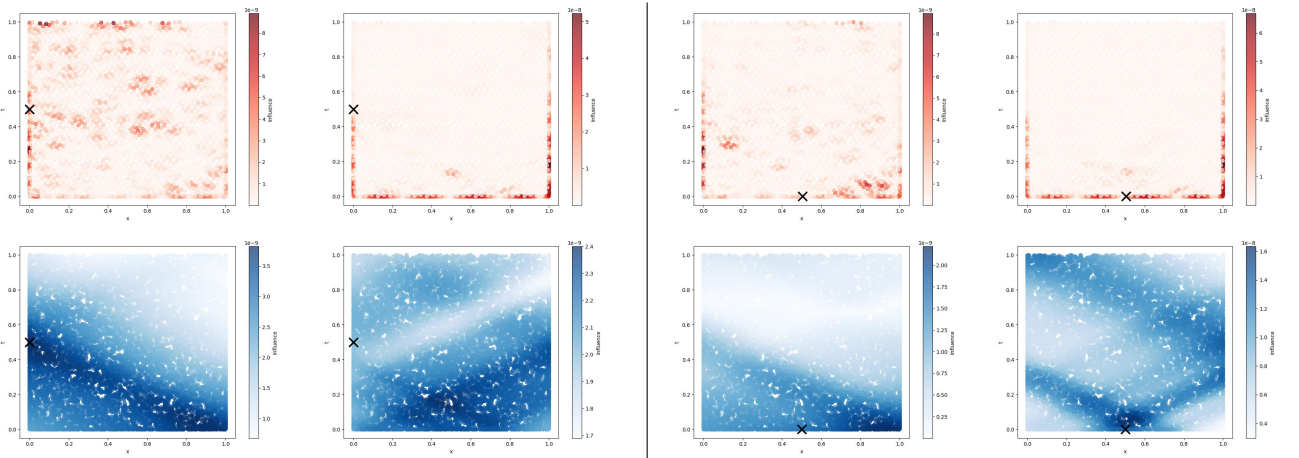


Figure 25. Wave equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

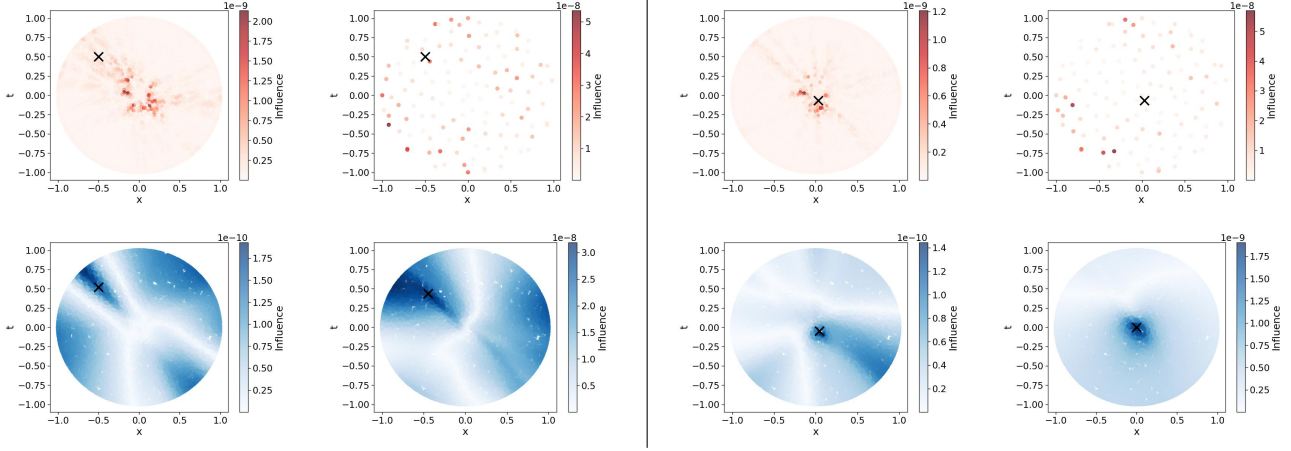


Figure 26. Poisson equation: Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

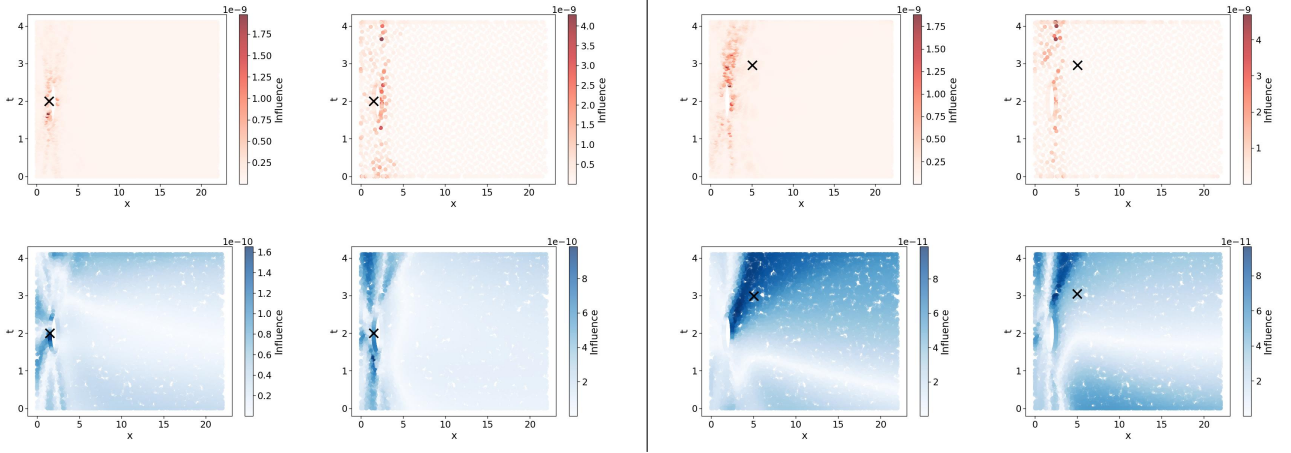


Figure 27. Navier-Stokes equations (x -Velocity): Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain

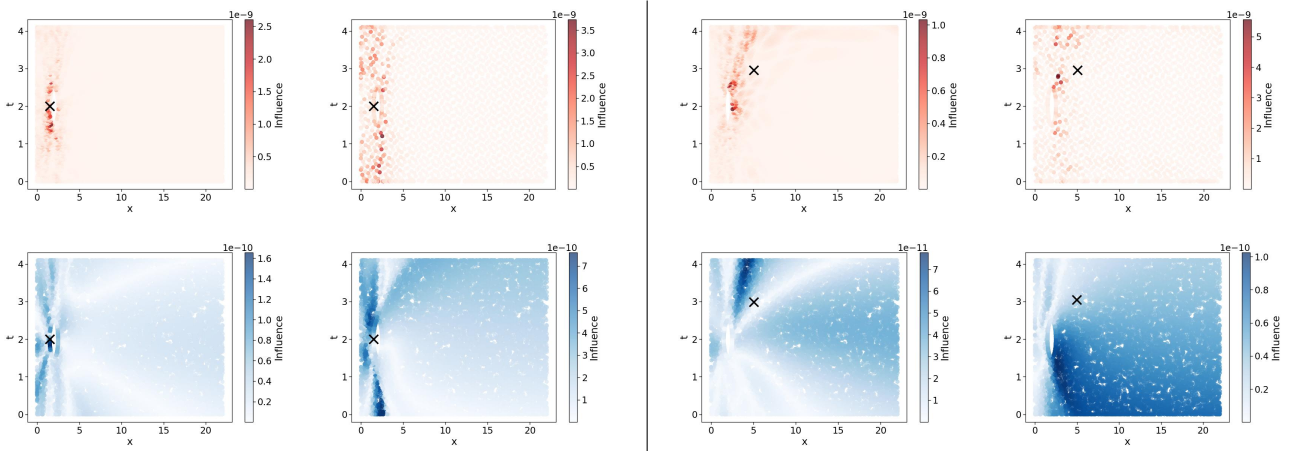


Figure 28. Navier-Stokes equations (y -Velocity): Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.

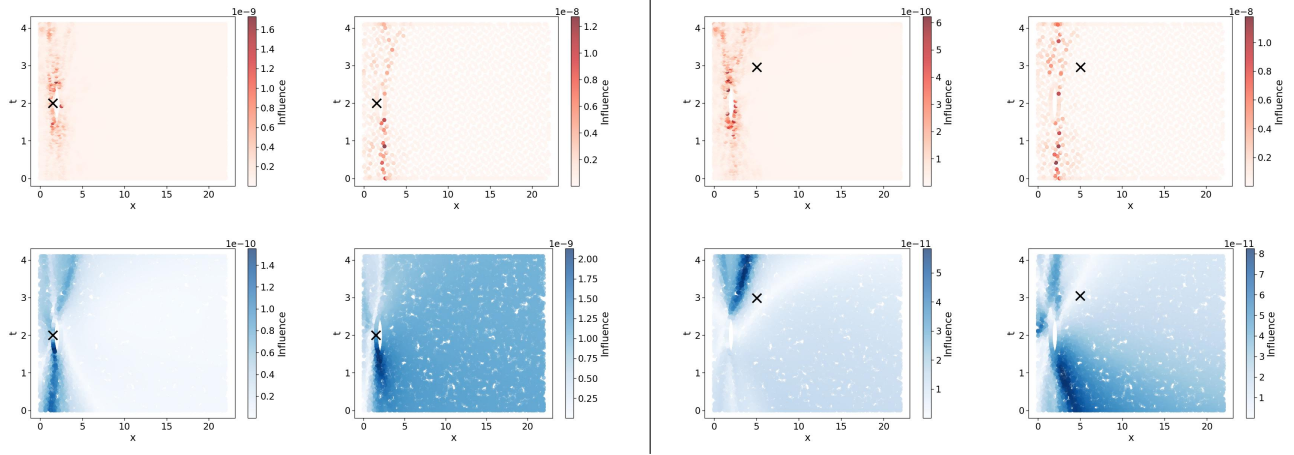


Figure 29. Navier-Stokes equations (Pressure). Influence scores for well-trained (left) and poorly-trained (right) models. Top: training point influences for fixed test point. Bottom: influences for a fixed training point onto the domain.